The Random Universe: a meeting to celebrate Andrew Jaffe at 60

Event | Conference

29 June 2026 - 2 July 2026

14.00 - 13.00 BST

Lecture Theatre 2, Blackett Building
South Kensington Campus

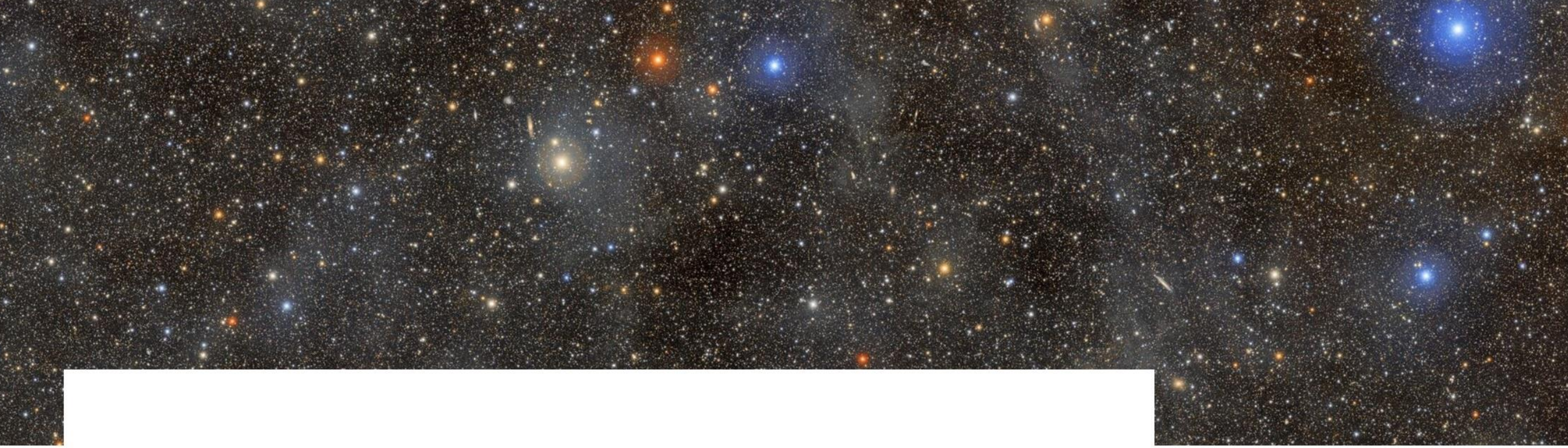
Accessibility information

Audience: Invitees only
Cost: Free
Tickets: Tickets to be purchased in advance

Register now

A conference to celebrate Andrew Jaffe's many contributions to cosmology and the exciting scientific developments still to come on his 60th birthday.

For his PhD advisor, this milestone is profoundly depressing.



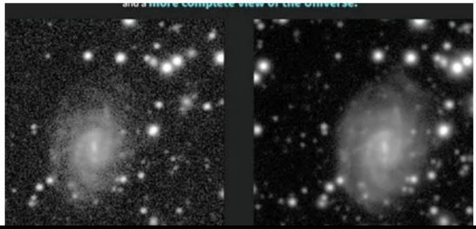
Action! NSF–DOE Vera C. Rubin Observatory Begins Capturing the Greatest Cosmic Movie Ever Made

The 10-year Legacy Survey of Space and Time has officially started, marking the beginning of a new era in astronomy and astrophysics

June 30, 2026

The wait is over: NSF–DOE Rubin Observatory, funded by the U.S. National Science Foundation and the U.S. Department of Energy's Office of Science, is now capturing the cosmos in unprecedented detail, transforming the way we study the dynamic Universe.

Media



Conceived when Andrew was a graduate student at Chicago

Is Dark Energy Evolving?

Josh Frieman
SLAC National Accelerator Laboratory
University of Chicago (emeritus)
Fermilab (emeritus)

The Random Universe:
A Meeting to Celebrate Andrew Jaffe at 60
July 1, 2026

IS HINCHLIFFE'S RULE TRUE? *

Boris Peon

Abstract

Hinchliffe has asserted that whenever the title of a paper is a question with a yes/no answer, the answer is always no. This paper demonstrates that Hinchliffe's assertion is false, but only if it is true.

Is This Article Consistent with Hinchliffe's Rule?

by Stuart M. Shieber, Professor of Computer Science
Harvard University, Cambridge, Massachusetts, USA

I demonstrate that Hinchliffe's rule—if the title of a scholarly article is a yes-no question, the answer is "no"—is paradoxical, by providing an article whose title is a question whose answer is "no" if and only if its answer is "yes".

Hinchliffe's Rule

Hinchliffe's rule, attributed with unknown veracity to Ian Hinchliffe (Holderness, 2014), is this: "If the title of a scholarly article is a yes-no question, the answer is 'no'." It can be seen as the academic analog of Betteridge's law of headlines (Betteridge, 2009).

In 1988, Boris Peon (which appears, thankfully if unsurprisingly, to be a pseudonym (Strassler, 2013)) distributed an article entitled "Is Hinchliffe's Rule True?" (Peon, 1988) Here is the abstract, which constitutes the article in its entirety:

> Hinchliffe has asserted that whenever the title of a paper is a question with a yes/no answer, the answer is always no. This paper demonstrates that Hinchliffe's assertion is false, but only if it is true.

Peon's article hints at Hinchliffe's rule being paradoxical, but it fails to manifest a true paradox. This is a lost opportunity, which I intend to correct by this very article.

The Problem with Peon's Purported Paradox

The second sentence of Peon's abstract asserts that Hinchliffe's rule is false only if it is true, intimating a connection to the liar paradox (Beall and Glanzberg, 2014). This assertion is false.

The construction "*p* if *q*" is an alternative to "if *q* then *p*". The converse, " if *p* then *q*" can be presented as "*p* only if *q*". (The conjunction of the two, "*p* if and only if *q*", therefore expresses the biconditional.) Peon's postulation can thus be reconstructed as stating "if Hinchliffe's rule is false then it is true".

But this does not follow from the existence of Peon's article. If Hinchliffe's rule is false, we can deduce nothing about the truth or falsity of that article's title, that is to say, nothing about the truth or falsity of Hinchliffe's rule, and certainly not that the rule is true. Only the other direction of reasoning holds, that if Hinchliffe's rule is true, then it is false: Suppose Hinchliffe's rule is true. Then the answer to the article's title ought to be "yes". But by Hinchliffe's rule, the answer to the article's title must be "no". The contradiction leads us to conclude that Hinchliffe's rule must be false, a conclusion that is self-consistent. (Of course, the same conclusion would follow from any article title in the form of a question with a positive answer, whether it mentions Hinchliffe's rule or not.) No paradox is exemplified by Peon's article.[1]

A Lost Opportunity Regained

Peon purports to demonstrate Hinchliffe's rule's paradoxical nature, but unfortunately has failed. The path to a paradox lies in the distinction between what the rule requires and what it is consistent with. Consider an article with a title question phrased as the negation of Peon's, viz., "Is Hinchliffe's Rule False?" If Hinchliffe's rule is true, then the answer to the title question is "no", which is consistent with the assumption, although it does not require it. If Hinchliffe's rule is false, then the answer to the title question is "yes", which is again consistent with the assumption, although it does not require it. We can conclude from such an article nothing about the truth of Hinchliffe's rule, but this case at least exhibits symmetry between the two horns of the dilemma, which will be needed in a paradoxical application of Hinchliffe's rule. Consistency with Hinchliffe's rule is the foundation on which to erect a paradox.

Taking a clue from this observation, consider a final example, an article—such as this very one—with the title "Is This Article Consistent with Hinchliffe's Rule?" Suppose the article is consistent with Hinchliffe's rule. Then the answer to the title question ought to be "no", that is, the article is not consistent with Hinchliffe's rule. Conversely, suppose the article is not consistent with Hinchliffe's rule. Then the answer to the title question must not be "no" (in other words, must be "yes"), stating that the article is consistent with Hinchliffe's rule. In either case, we have a contradiction, and Hinchliffe's rule leads to a paradox.

At least now that this article has been published.

Note

1. Perhaps the conditional in Peon's abstract was intended to be interpreted as encompassing its converse as well via so-called "conditional perfection" (Horn, 2000), the kind of reasoning that allows moving from "I'll give you five bucks if you shut up" to the implicated condition "I'll give you five bucks *only if* you shut up". But this strengthened interpretation as a biconditional ("Hinchliffe's rule is false if and only if it is true") is also not paradoxical, merely false.

Andrew as a graduate student at UChicago

- Cosmological Constraints on pseudo-Nambu Goldstone Bosons
  Jaffe+JF 1992 (ahead of its time: foreshadowed ALPs)
- Limits on Neutrino Radiative Decay from SN1987A
  Jaffe, Fenimore, and Turner 1993
- Minimal Microwave Anisotropy from Perturbations Induced at Late Times Jaffe, Stebbins, JF 1994
- Quasi-linear Evolution of Compensated Cosmological Perturbations: the Non-linear Sigma Model, PhD thesis, 1994

- Rock criticism in the Chicago free newspaper *New City*:

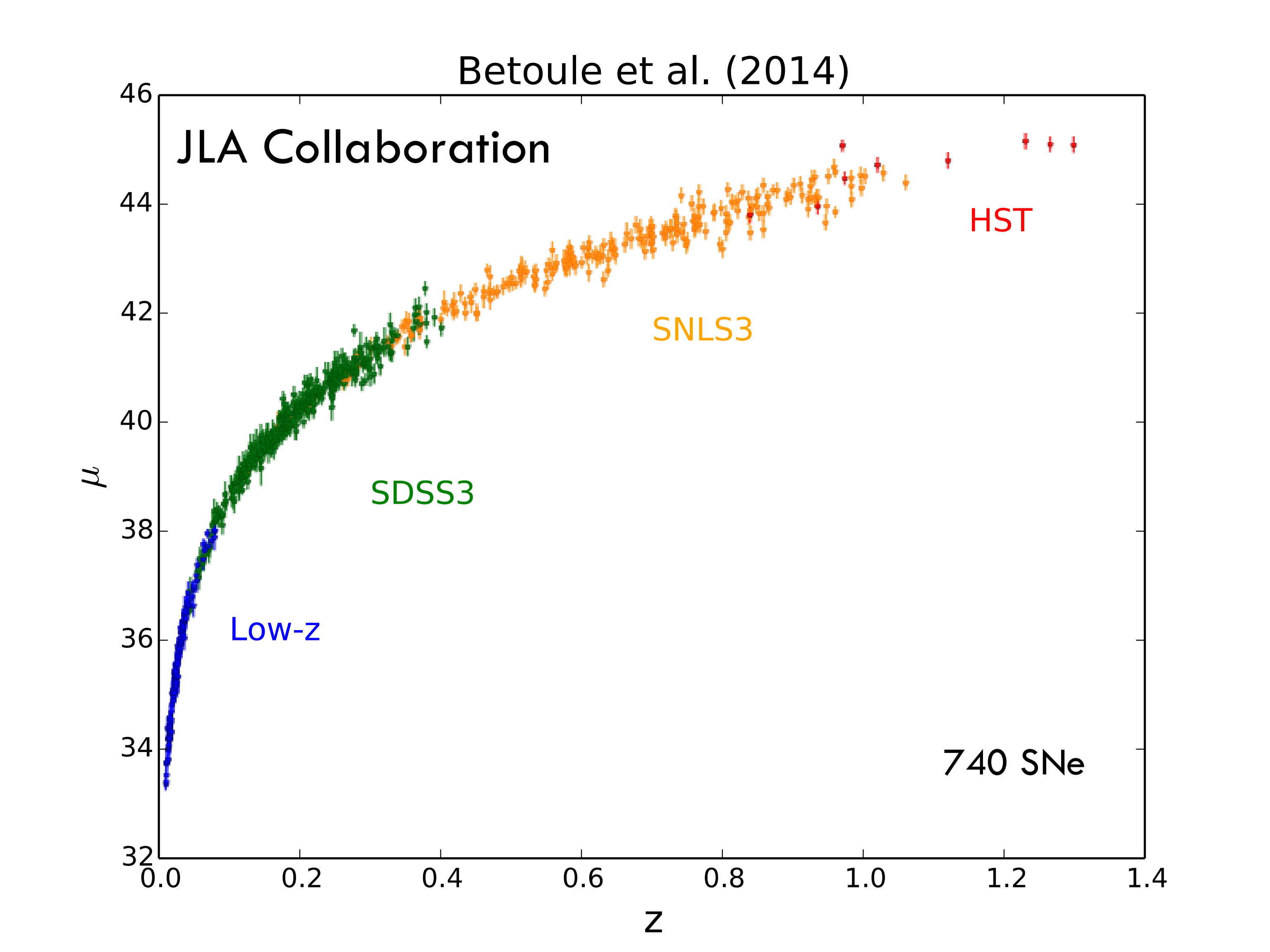
"Nirvana recently did an all-ages show here in Chicago, which I guess means I'm supposed to be too old to like this sludge. Unlike the rest of their Sub Pop labelmates (although Nevermind is being released as a joint venture with Geffen's DGC), Nirvana is less a continuation of the boring revalorization of Led Zeppelin than a vision of Hollywood-style popmetal from Poison to the Crüe redone by ... In fact, their obvious lack of technique pegs them as indie rockers rather than metalheads, if such distinctions matter…"

alas, if only physics articles were written in this style…

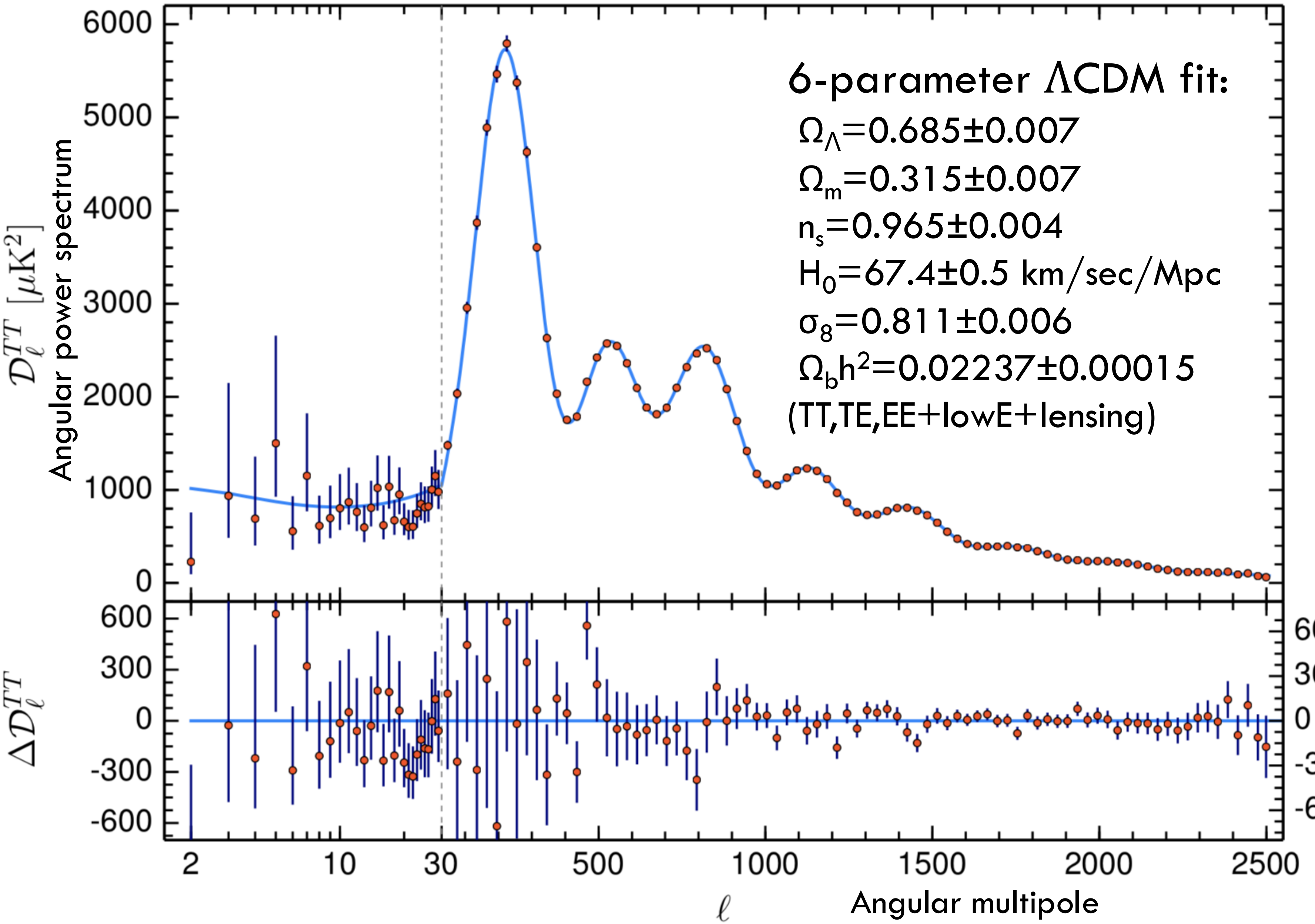
What is the Nature of Dark Energy?

- Despite remarkable success of the ΛCDM paradigm, we don't understand the *physics* of dark energy:
  - Cosmological constant Λ?
  - Or a *dynamical, evolving* phenomenon?
    - Is DE equation of state parameter $w = -1$ or not?
  - Or a modification of General Relativity?
- Moreover, in the last ~2.5 years, cracks have started to appear in ΛCDM, rekindling interest in alternatives to Λ.
- Present and near-future cosmic surveys are/will stress-test, ΛCDM by constraining cosmological parameters at the sub-percent level.

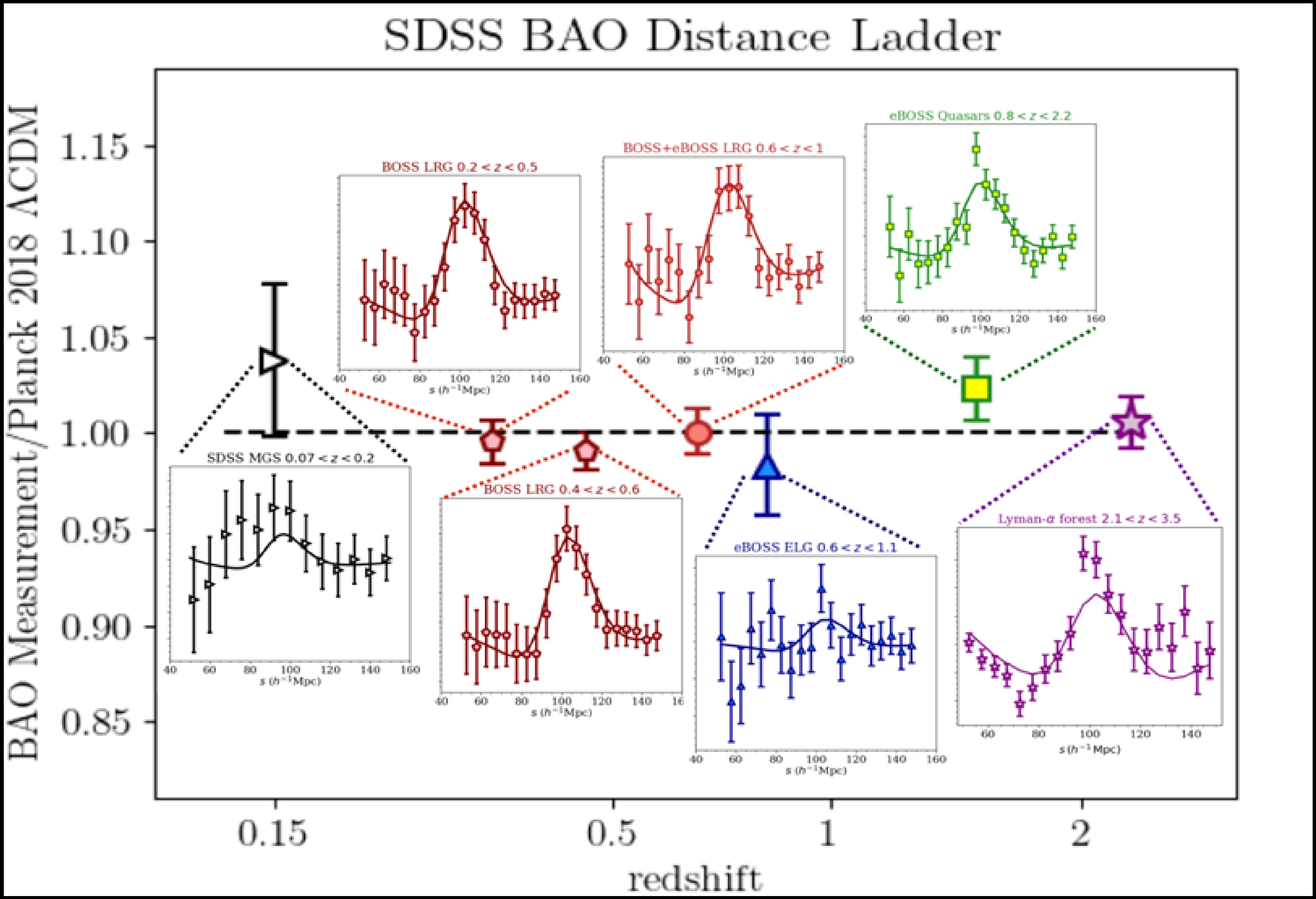
JLA SN Ia Hubble Diagram



Planck 2018 CMB Results

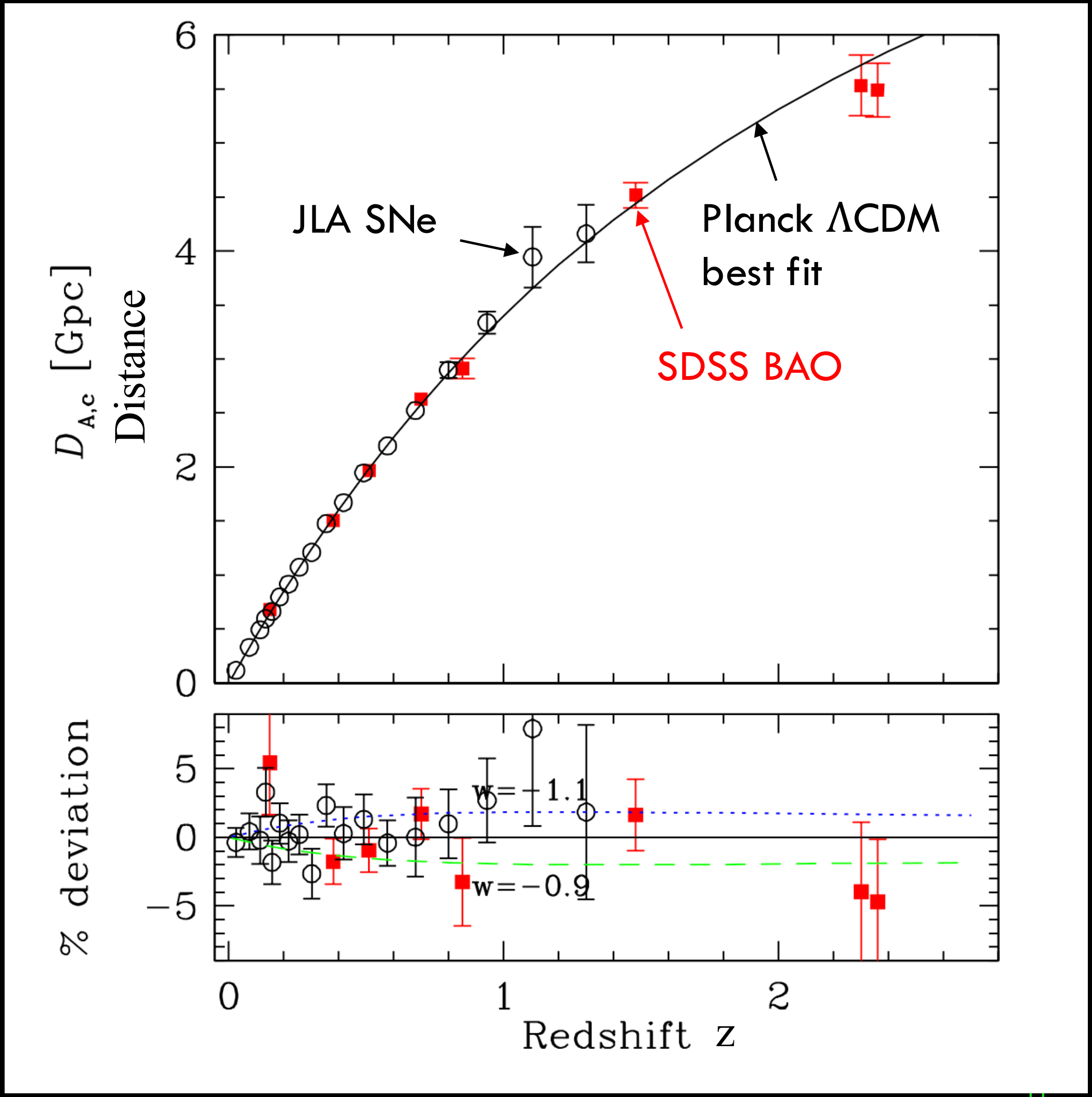


SDSS Baryon Acoustic Oscillations



SDSS: Alam+ 2021

ΛCDM Cosmic Concordance c. 2022



Weinberg and White,
Particle Data Group, 2022

ΛCDM Cosmic Concordance 2022

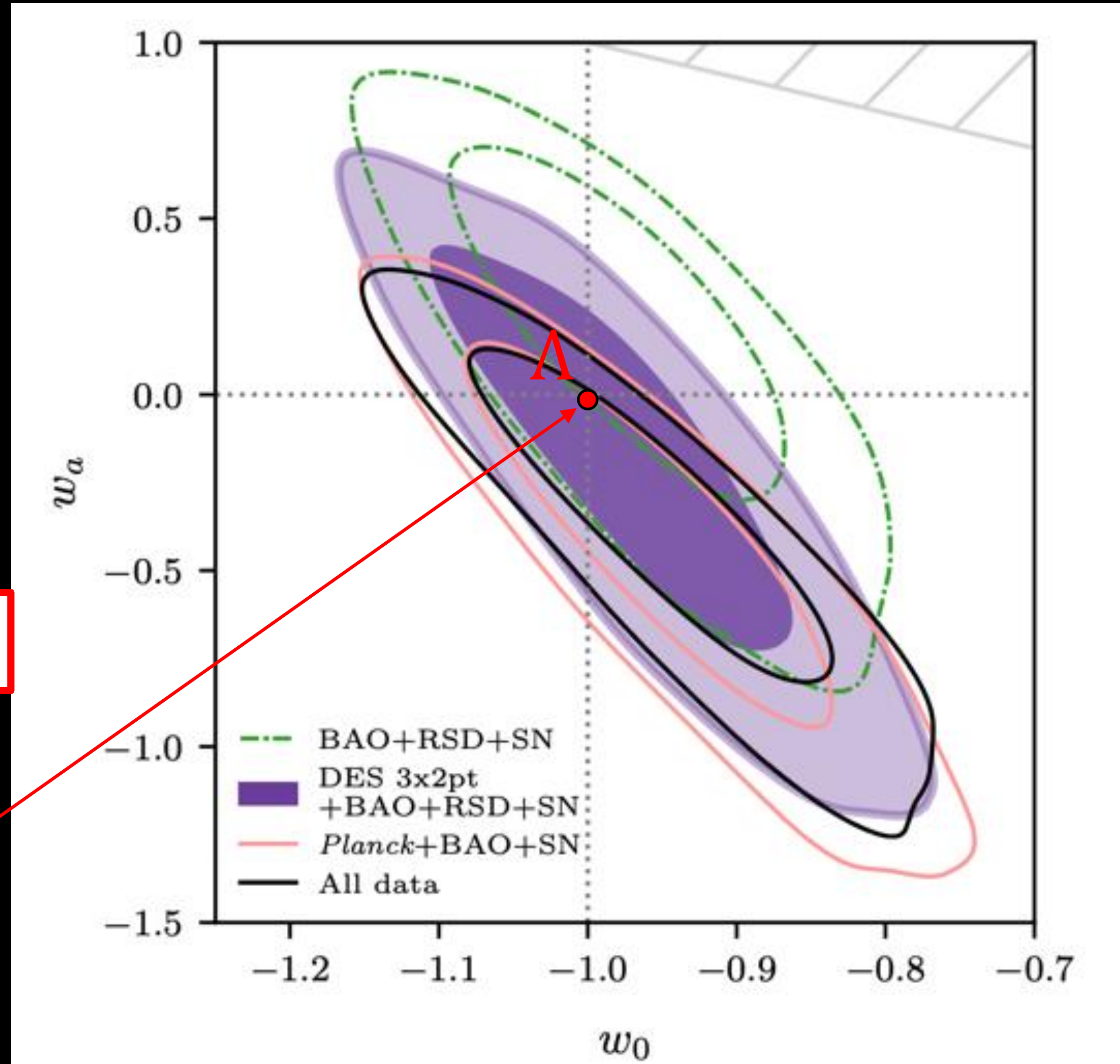
Parameterize Evolving Dark Energy model by:

$$w(a) = w_o + (1-a)w_a$$

$$w_0 = -0.95 \pm 0.08, \quad w_a = -0.4^{+0.4}_{-0.3}$$

consistent with ΛCDM

$$(w_0 = -1, w_a = 0)$$



DES Y3 3x2 + Pantheon SN + SDSS BAO + CMB

DES Collaboration 2022

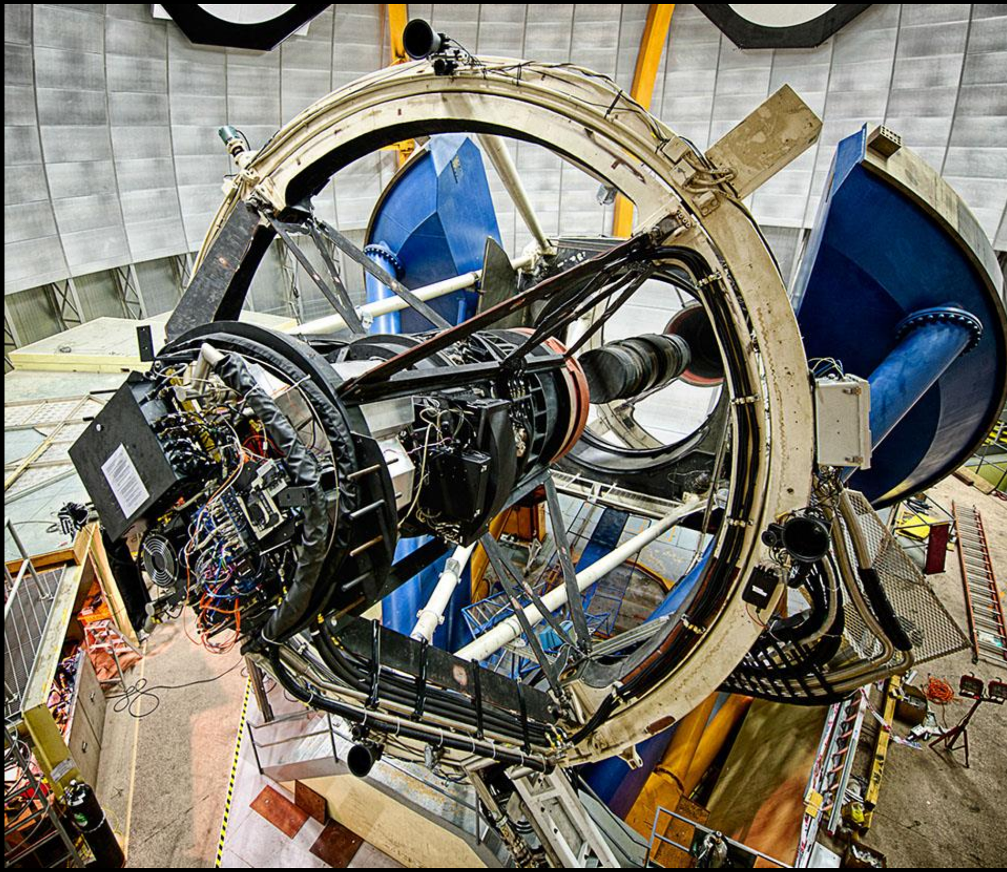
2023: All was well with ΛCDM.
What's changed in the last ~2.5 years?

- Improved Expansion History Constraints:
  - New Supernova Ia Results from DES, Union Jan. 2024, Nov. 2025
  - New BAO Results from DESI (DR1, DR2) and DES (DES-Y6)
    April '24, Mar. '25 Mar. '25
  - Combined SN Ia Results from Pantheon+ and DES
    July '26
- Improved Growth of Structure Constraints:
  - DES Year-6 Weak Lensing+Galaxy Clustering (3x2pt) Jan., May '26
    - Deeper data, multiple analysis improvements compared to DES Y3

Dark Energy Survey (DES)

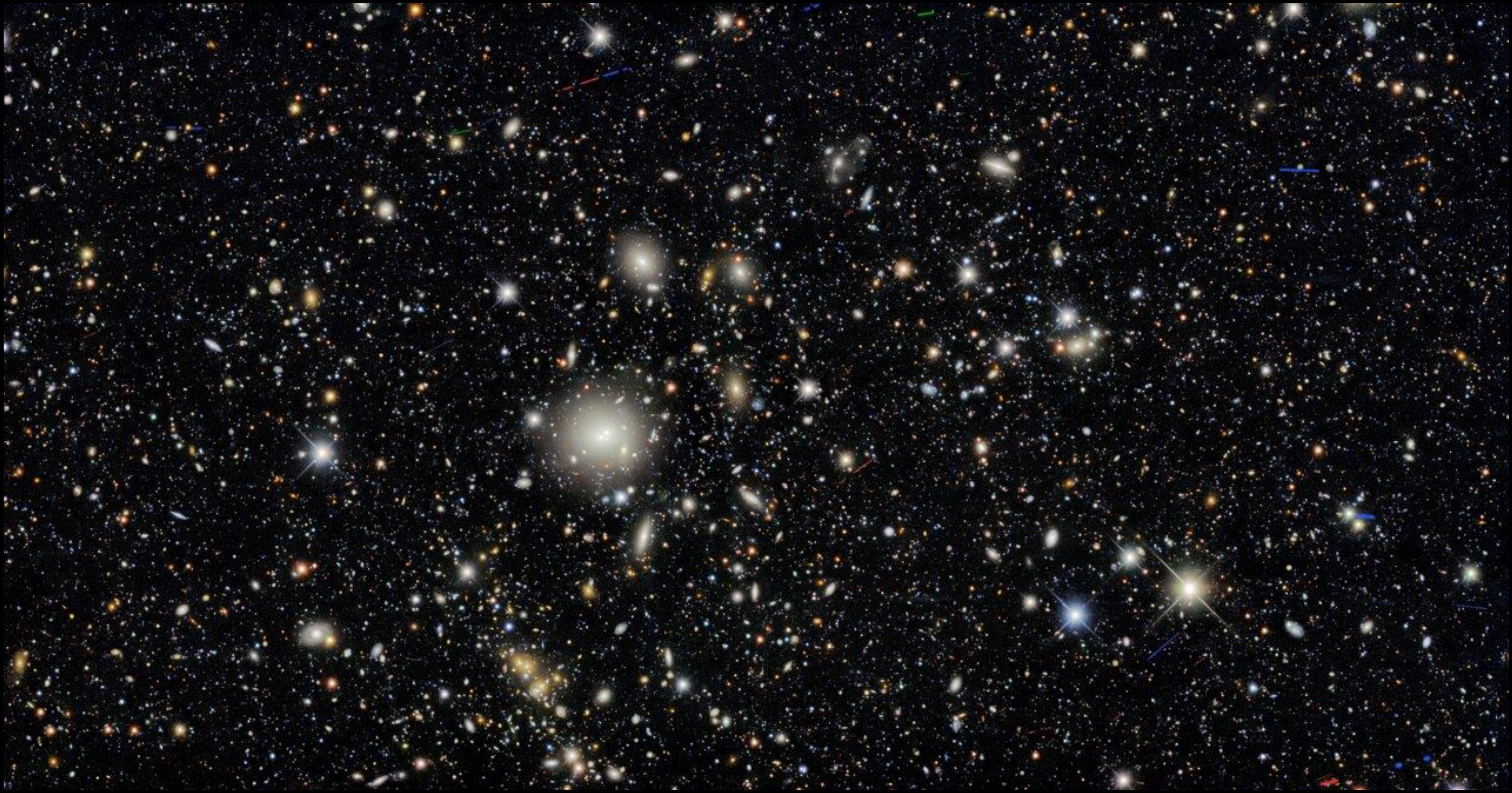
DECam on the CTIO Blanco 4m

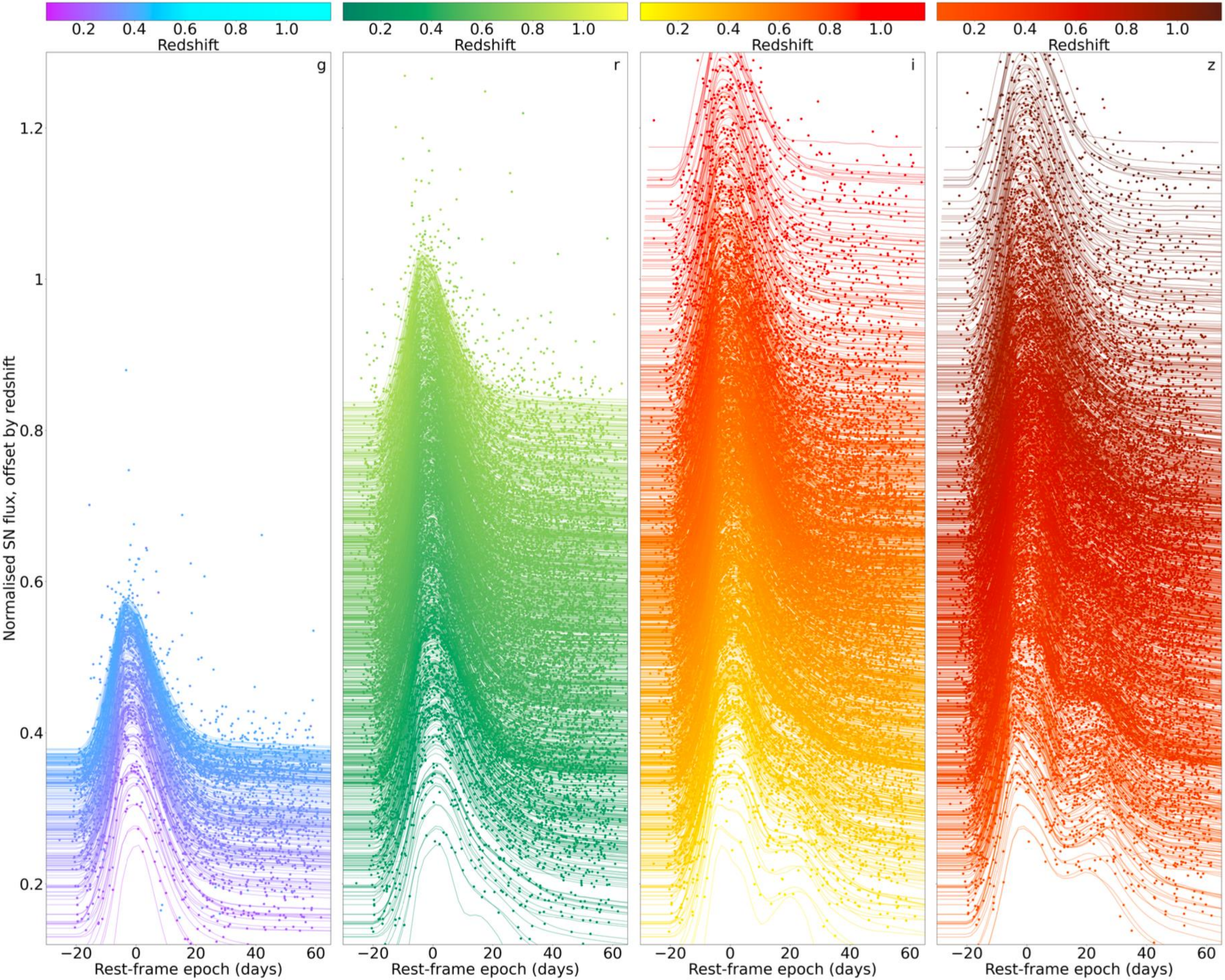
- Probe Dark Energy via:
  - Clusters, Weak Lensing, Galaxy clustering, Supernovae
- Two multicolor surveys:
  - 200M+ galaxies over 1/8 sky
  - 2000 supernovae (27 sq deg)
- *570 Megapixel Camera*
  - Built at Fermilab
- Survey Aug. 2013-Jan. 2019
  - Final data analyses about complete
  - Critical contributions from DES:UK groups



International collaboration of ~400+ scientists from 7 countries

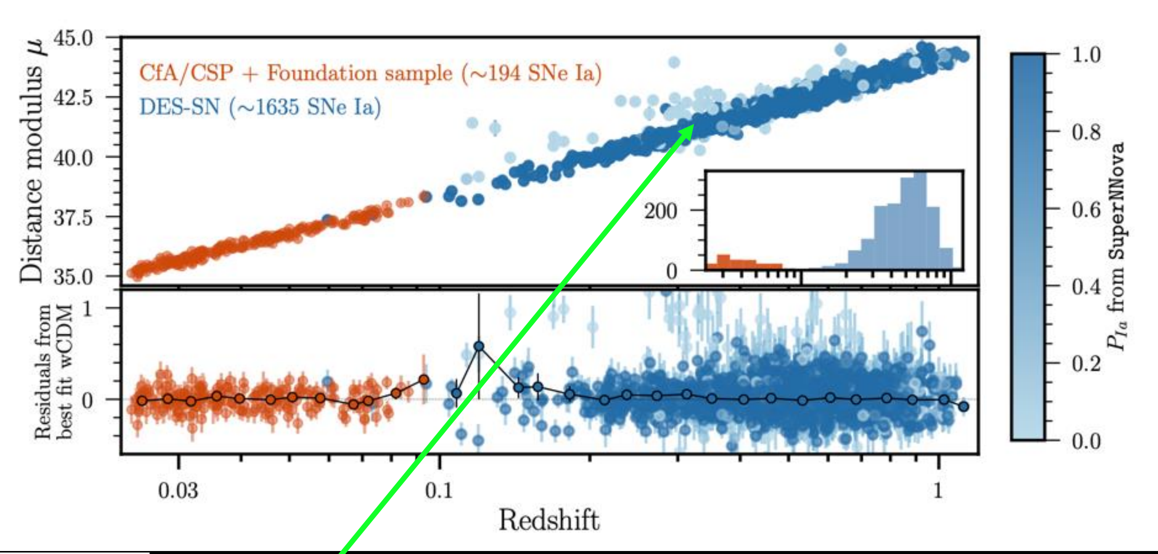
DES Deep SN Field Image

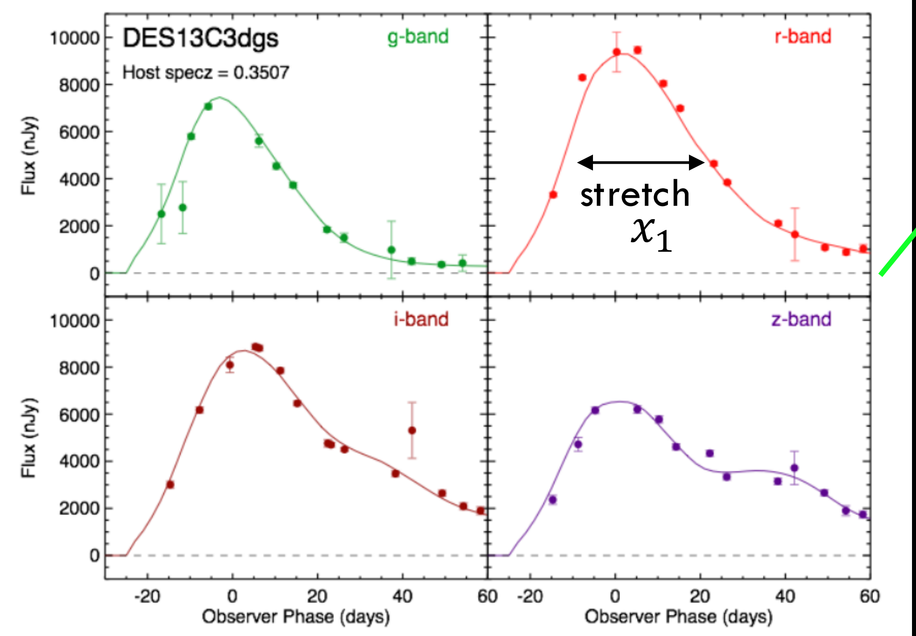




0.2 0.4 0.6 0.8 1.0
Redshift
g
r
i
z
1.2
1
0.8
0.6
0.4
0.2
Normalised SN flux, offset by redshift
−20 0 20 40 60
Rest-frame epoch (days)

DES 5-Year Supernova Ia Cosmology





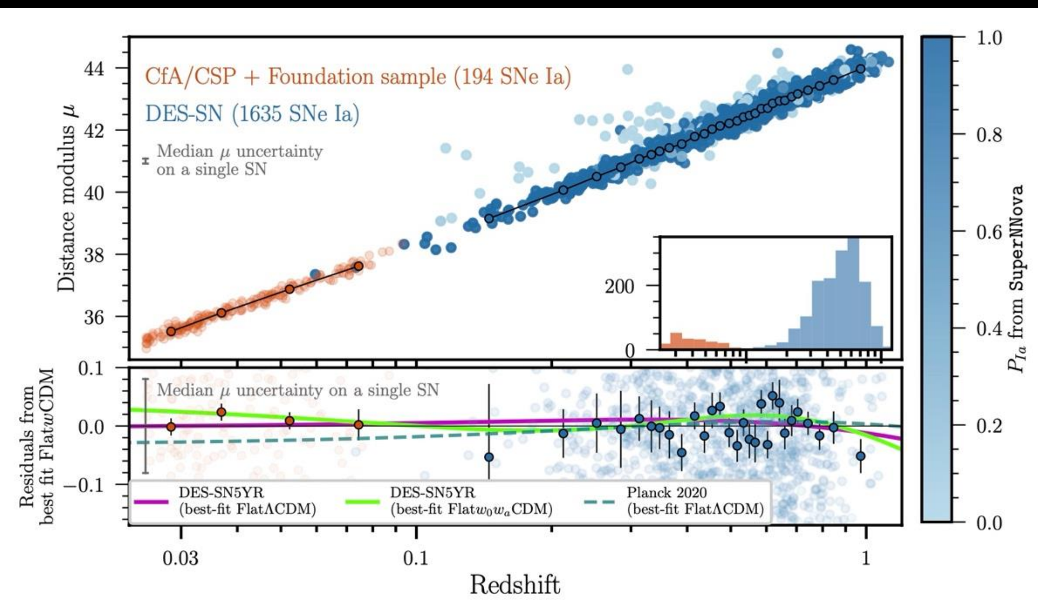
Vincenzi etal 2023
DES Collaboration 2024

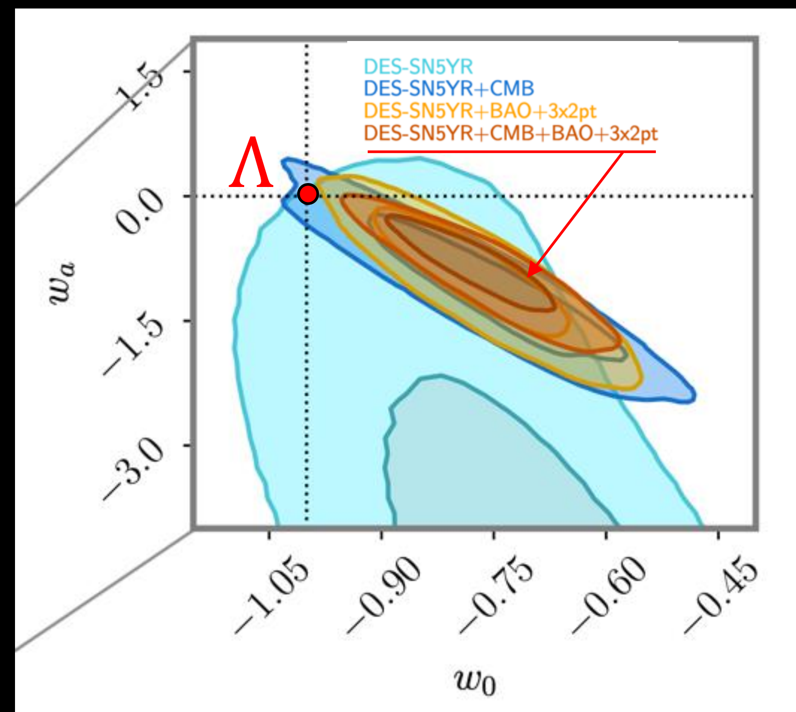
SN Ia distance estimated from peak brightness, light-curve stretch & color (empirical trained linear fit) with ~7% precision:

$$\mu_{\text{obs},i} = m_{x,i} + \alpha x_{1,i} - \beta c_i + \gamma G_{\text{host},i} - M - \Delta\mu_{\text{bias},i}$$

DARK ENERGY SURVEY

DES 5-Year Supernova Results





DES-SN5YR+
SDSS(Pre-DESI) BAO
+CMB+DESY3 3x2pt

Combined constraints:

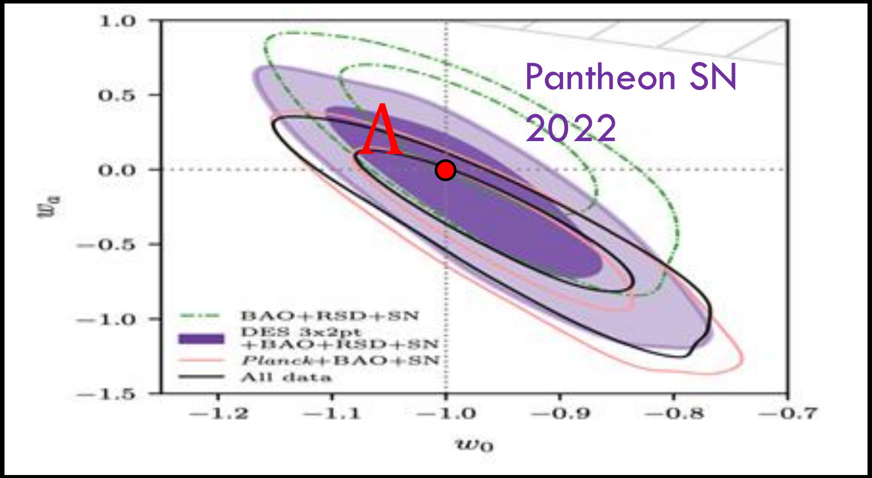
$$w_0 = -0.773^{+0.075}_{-0.067}$$
$$w_a = -0.83^{+0.33}_{-0.42}$$

~2.5σ from ΛCDM

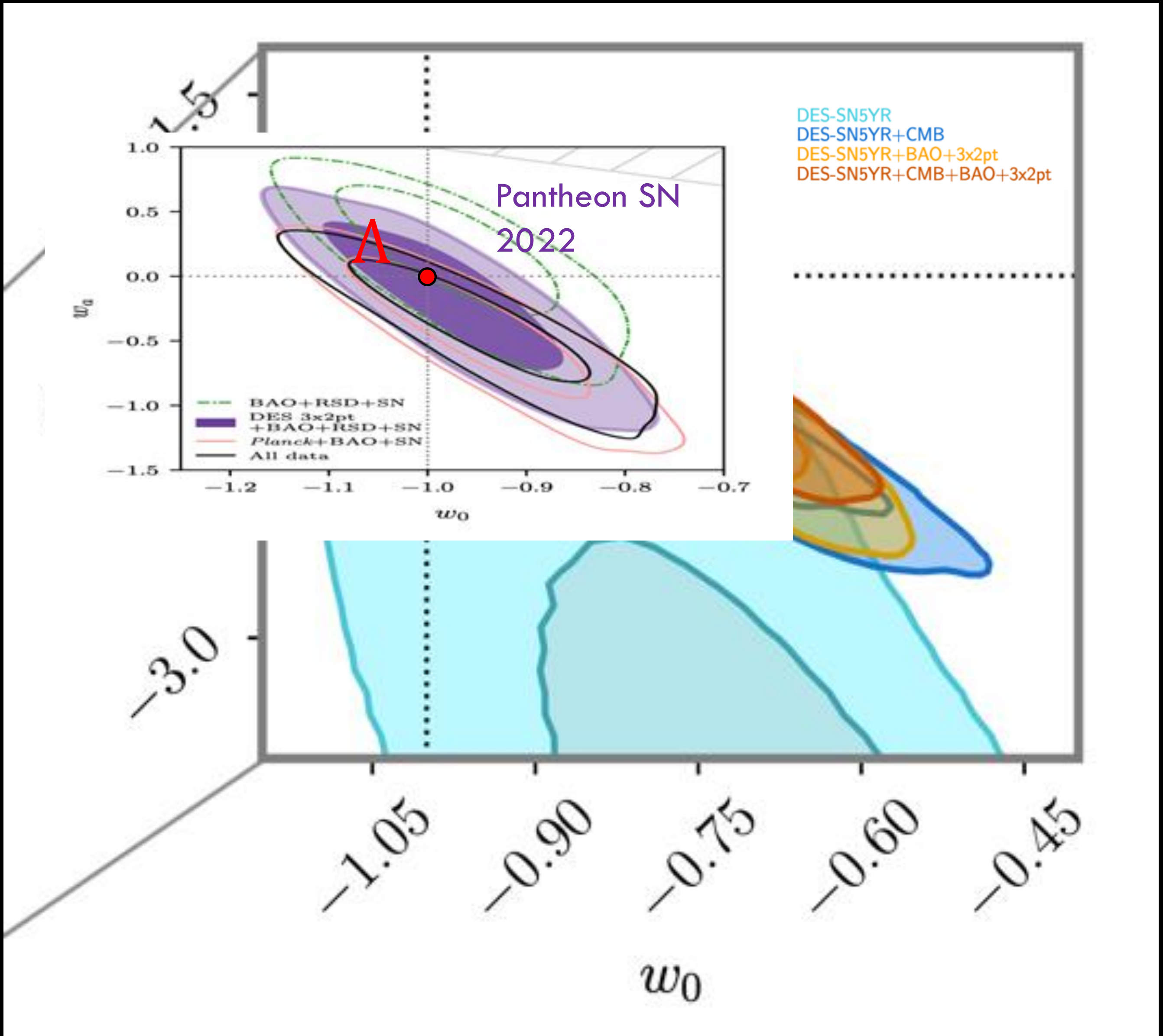
DES Collaboration, Jan. 2024

$$w(a) = w_o + (1-a)w_a$$
$$w(z) = w_0 + \left(\frac{z}{1+z}\right) w_a$$

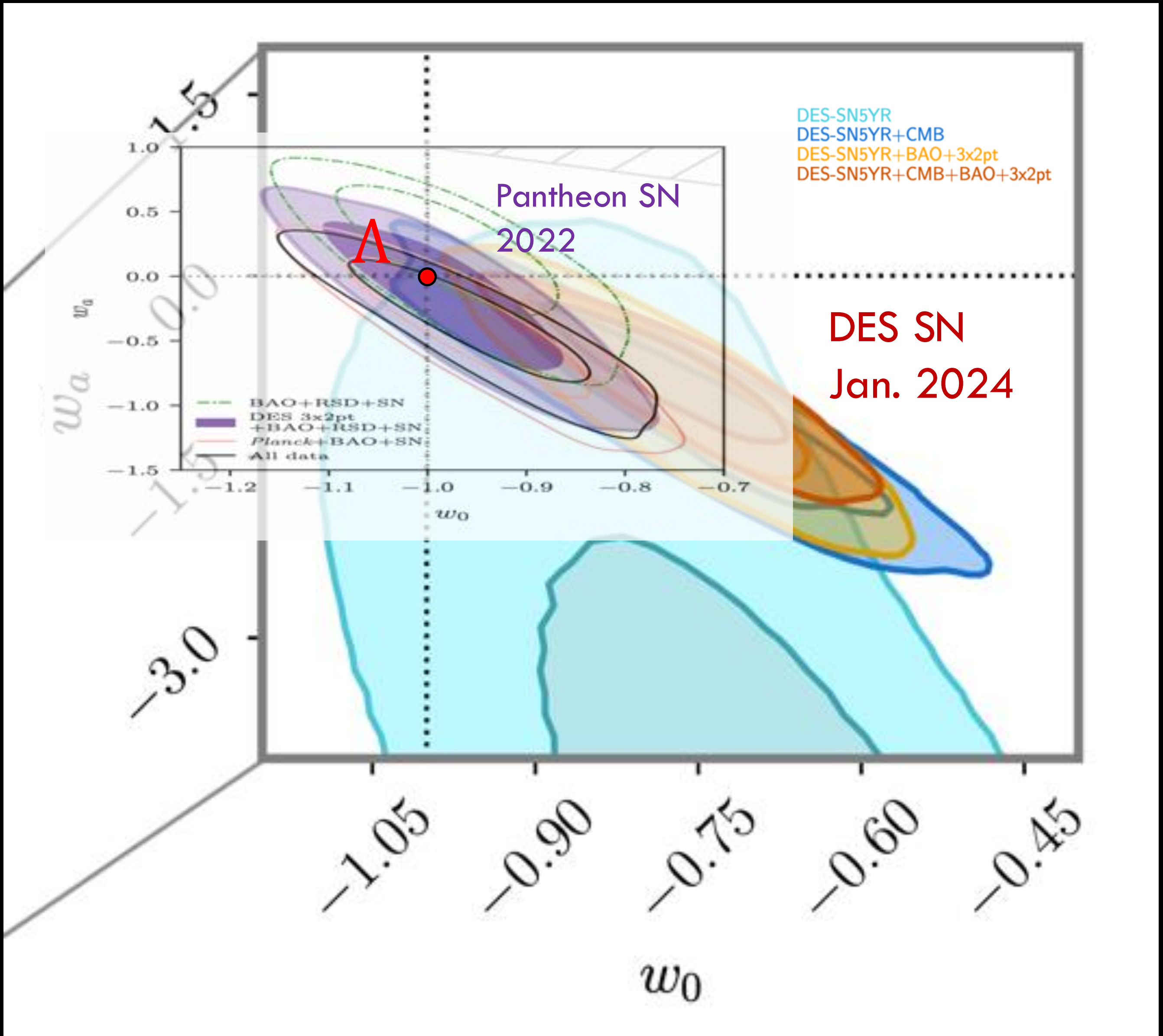
Supernovae: Pantheon



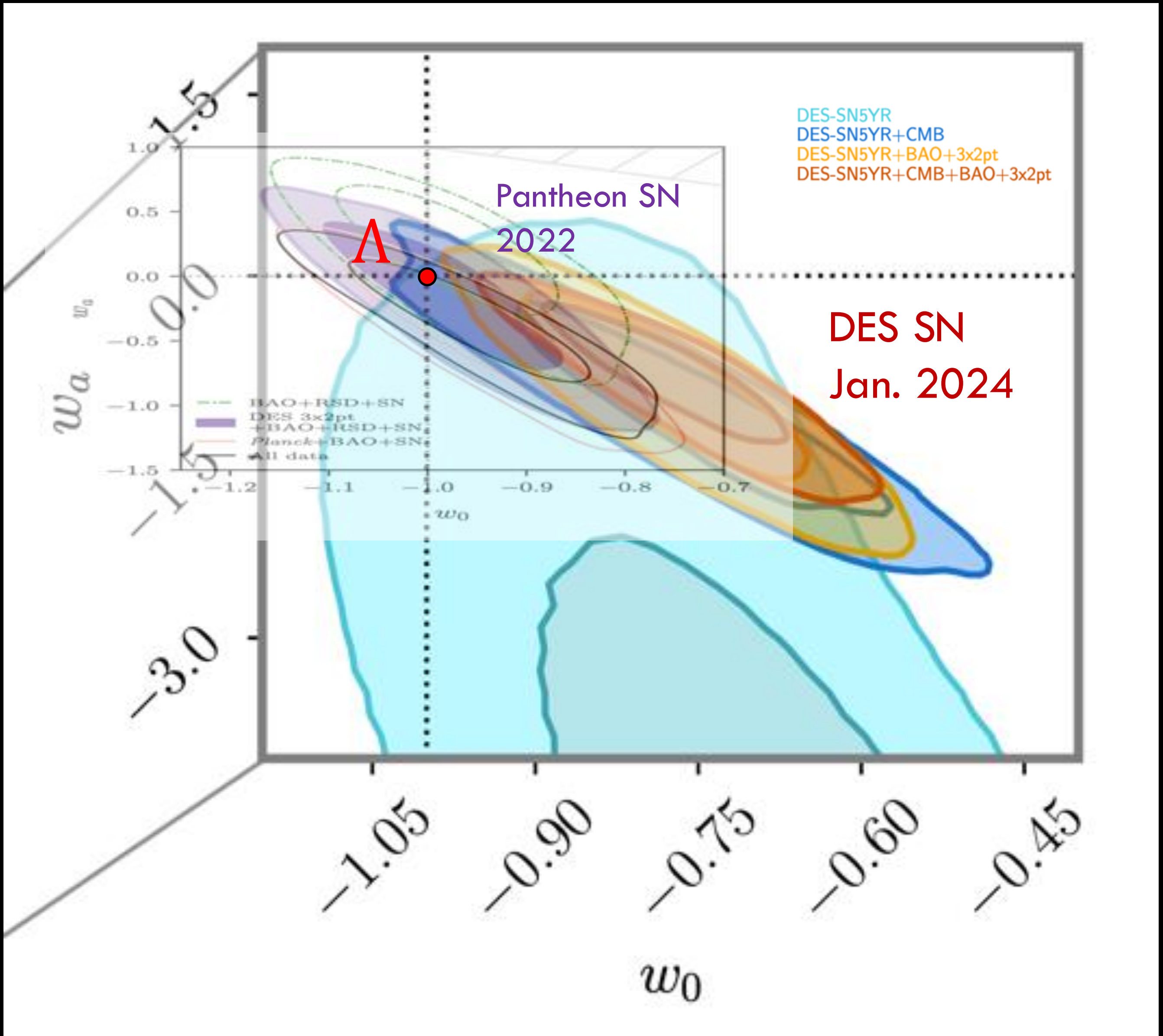
Supernovae: Pantheon → DES SN



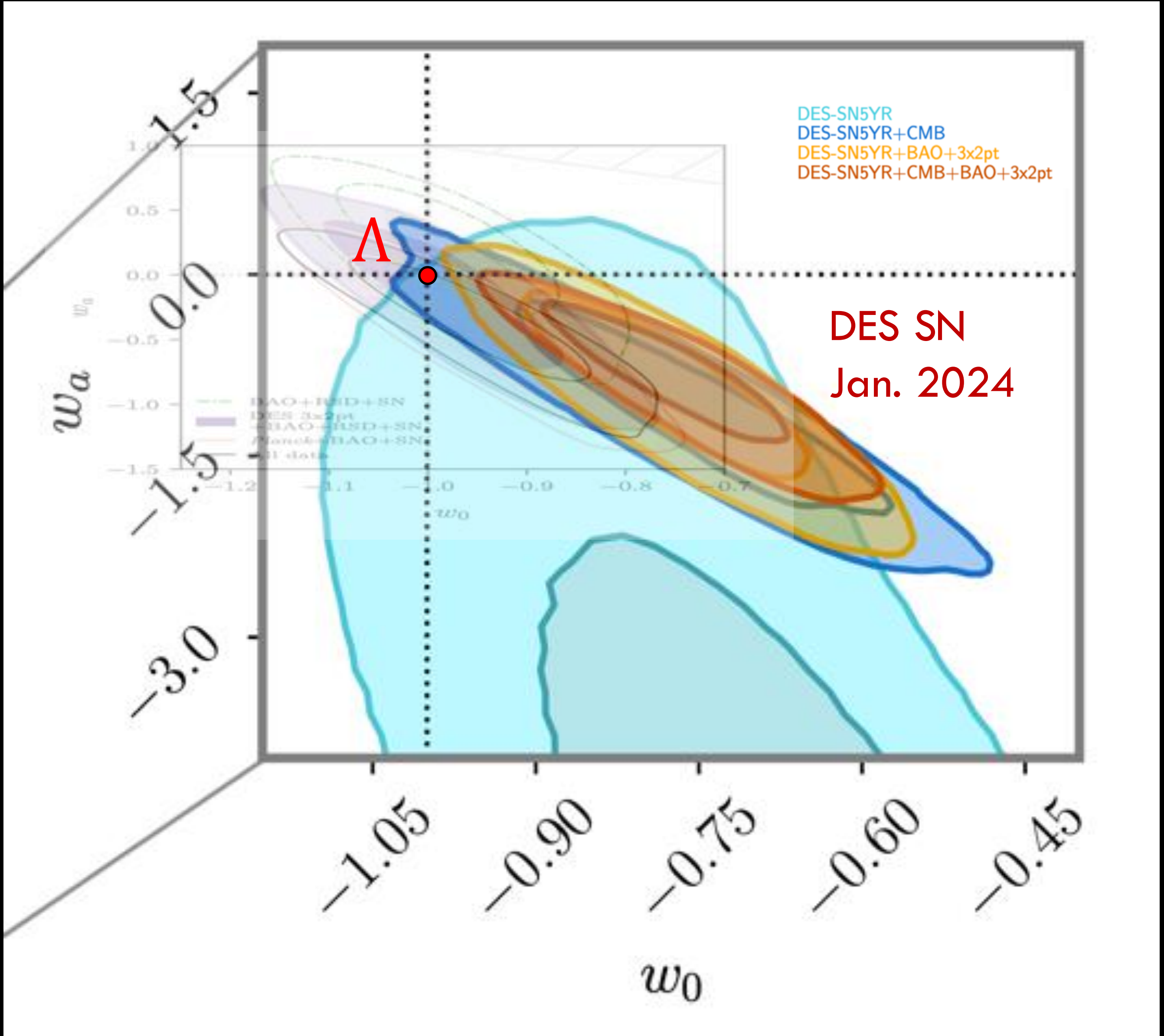
Supernovae: Pantheon → DES SN



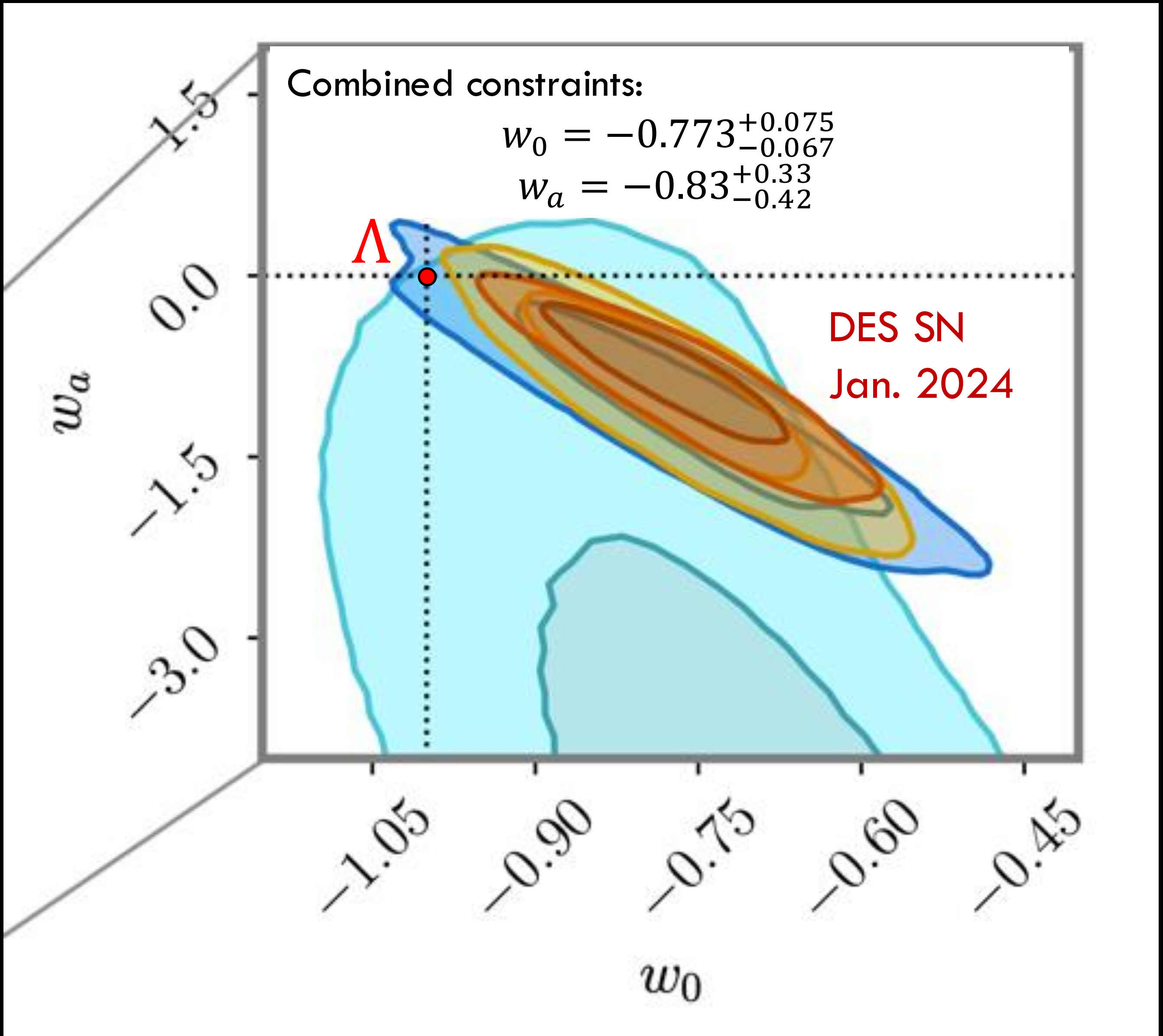
Supernovae: Pantheon → DES SN



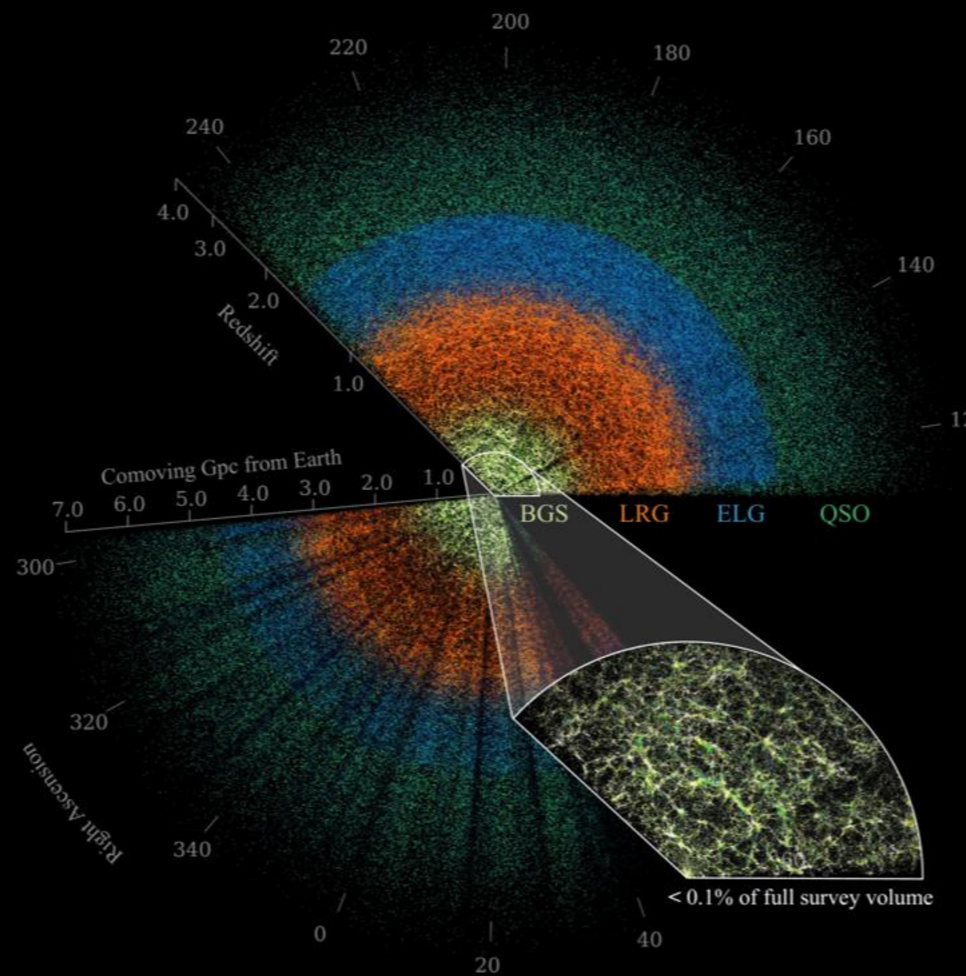
Supernovae: Pantheon → DES SN

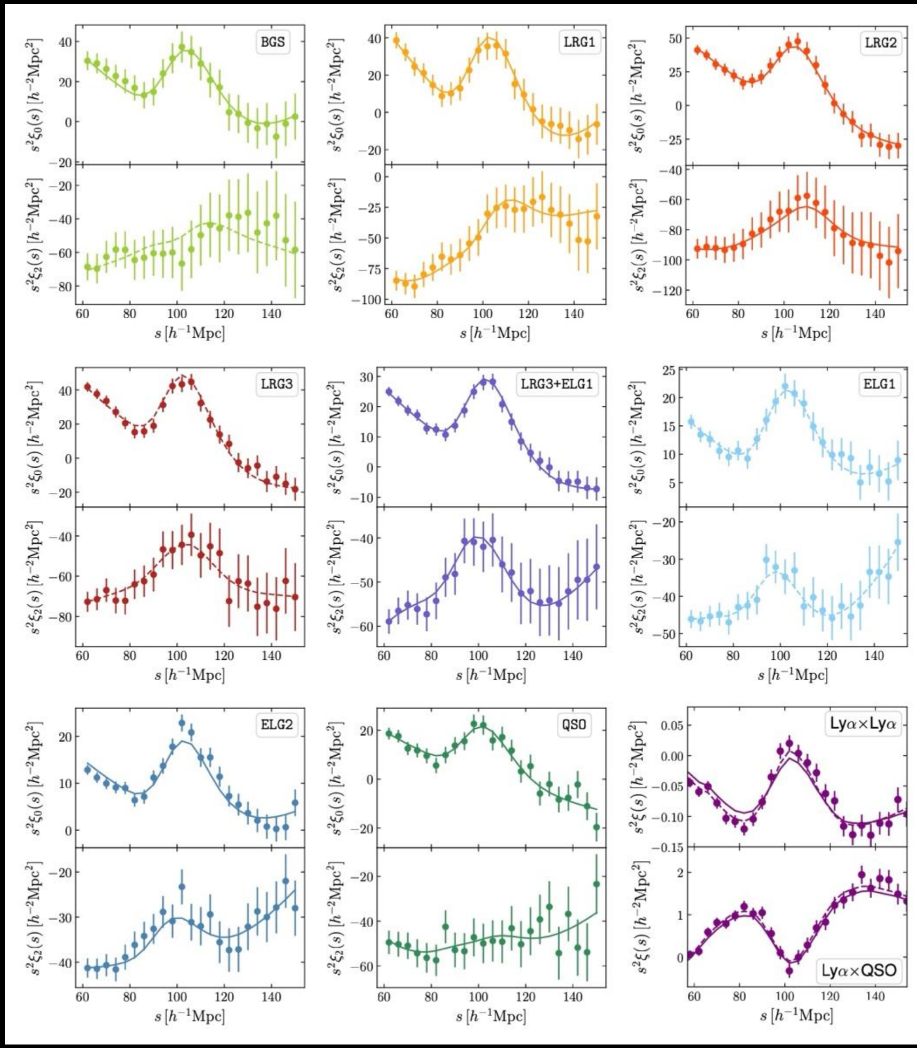


Supernovae: Pantheon → DES SN



DESI DR2 BAO: March 2025

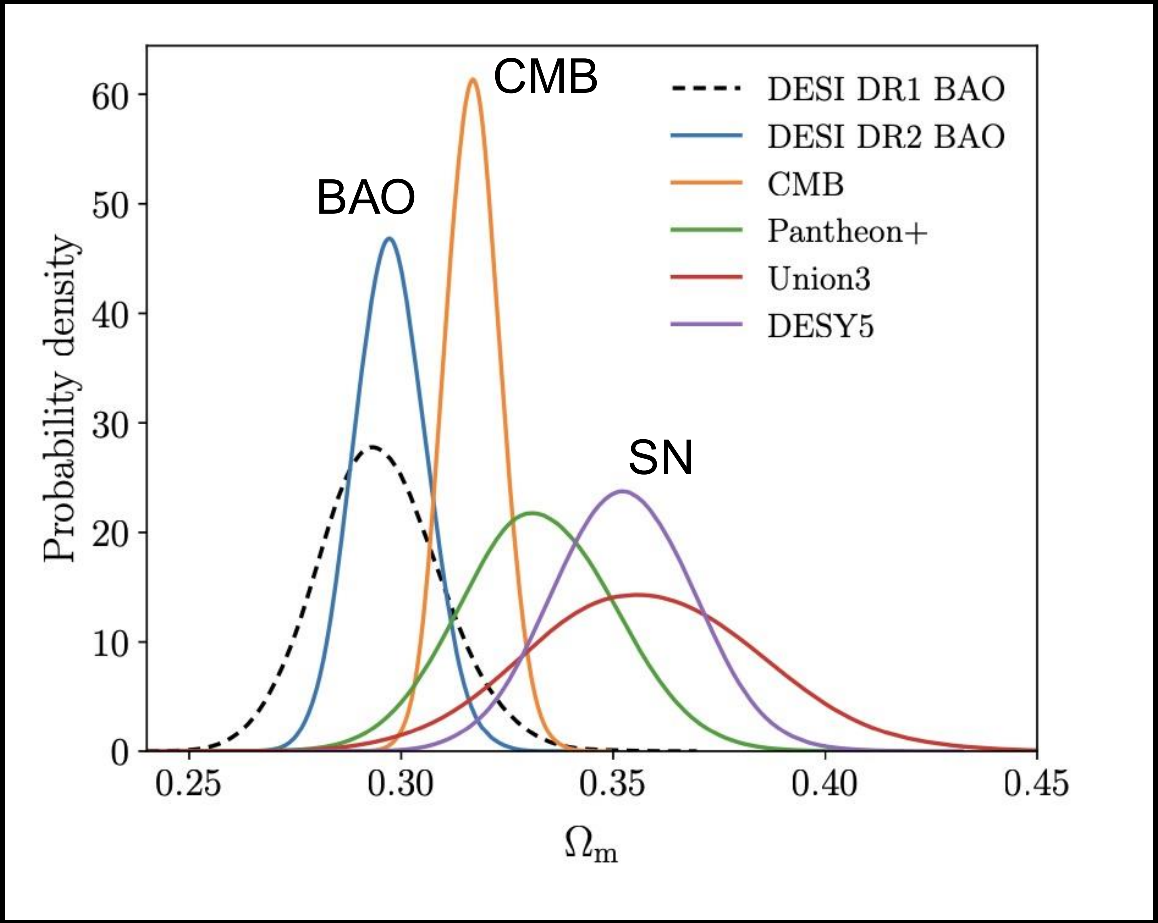




DESI DR2 (Y3), 2025

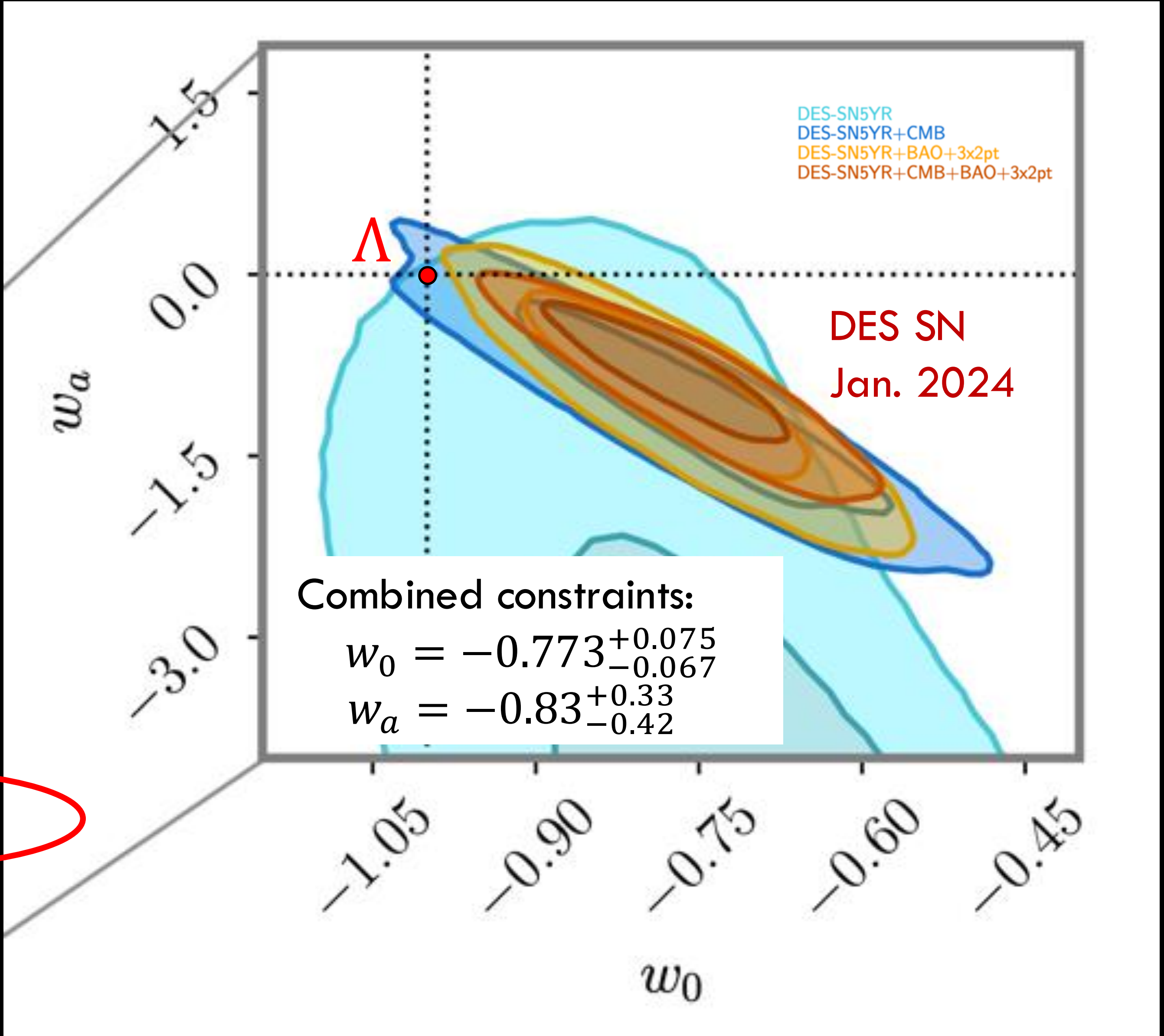
ΛCDM: DESI DR2 BAO vs CMB, SN

Tension: the *same* ΛCDM model doesn't fit all the data.



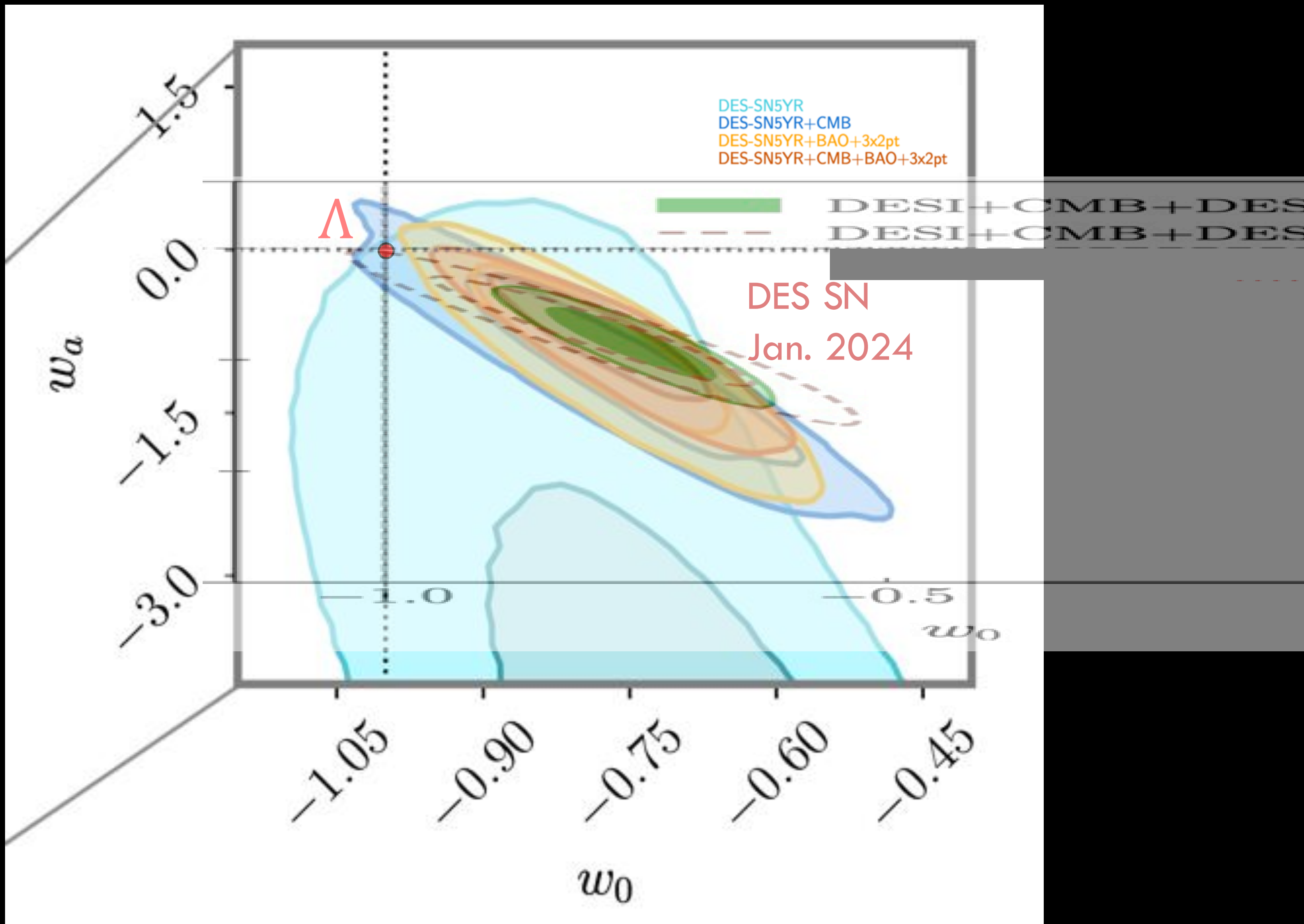
DESI DR2, 2025
See also DES 2025

DES SN+ SDSS BAO

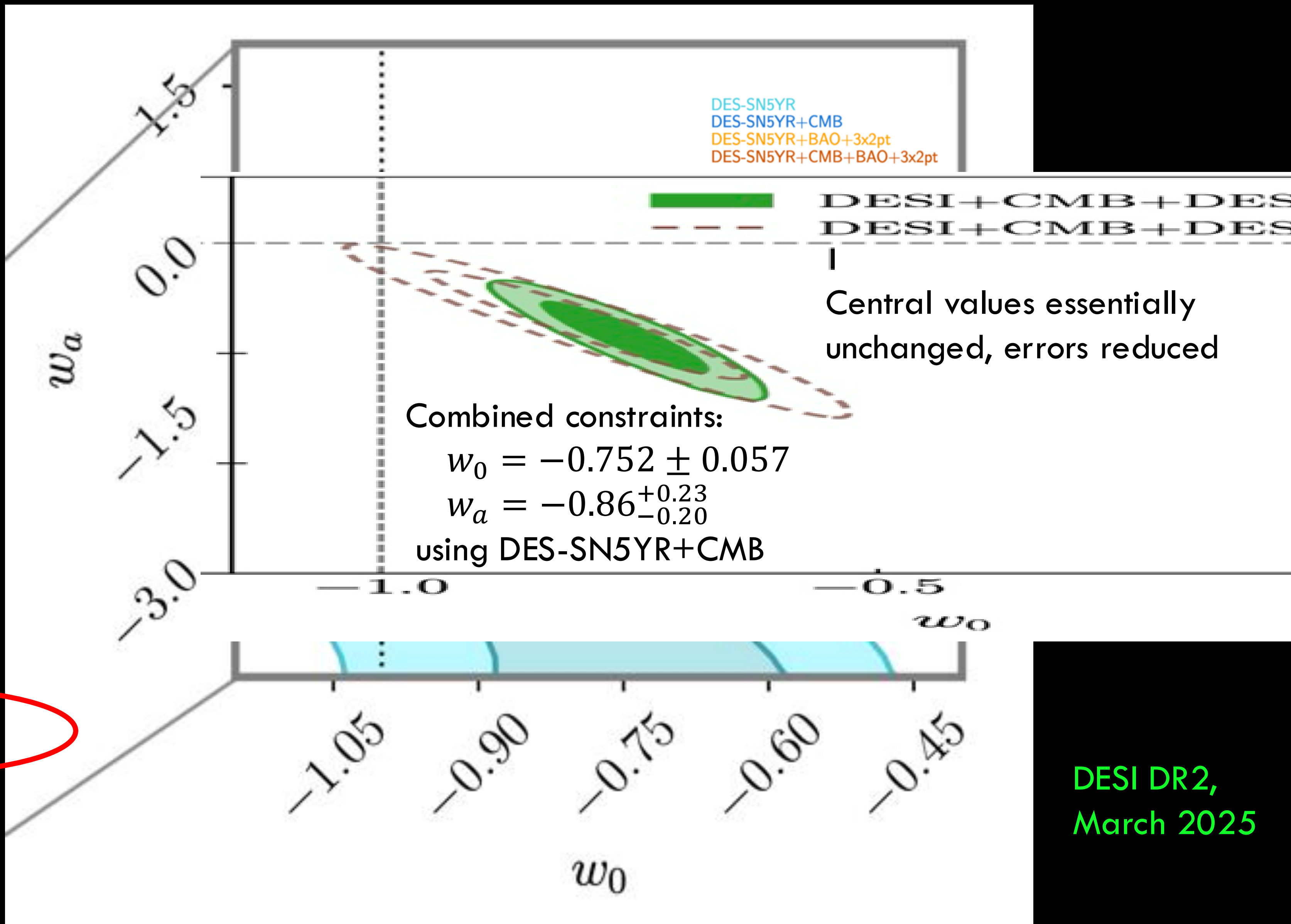


$\sim 2.5\sigma$ from ΛCDM

DES SN+DESI DR2 BAO



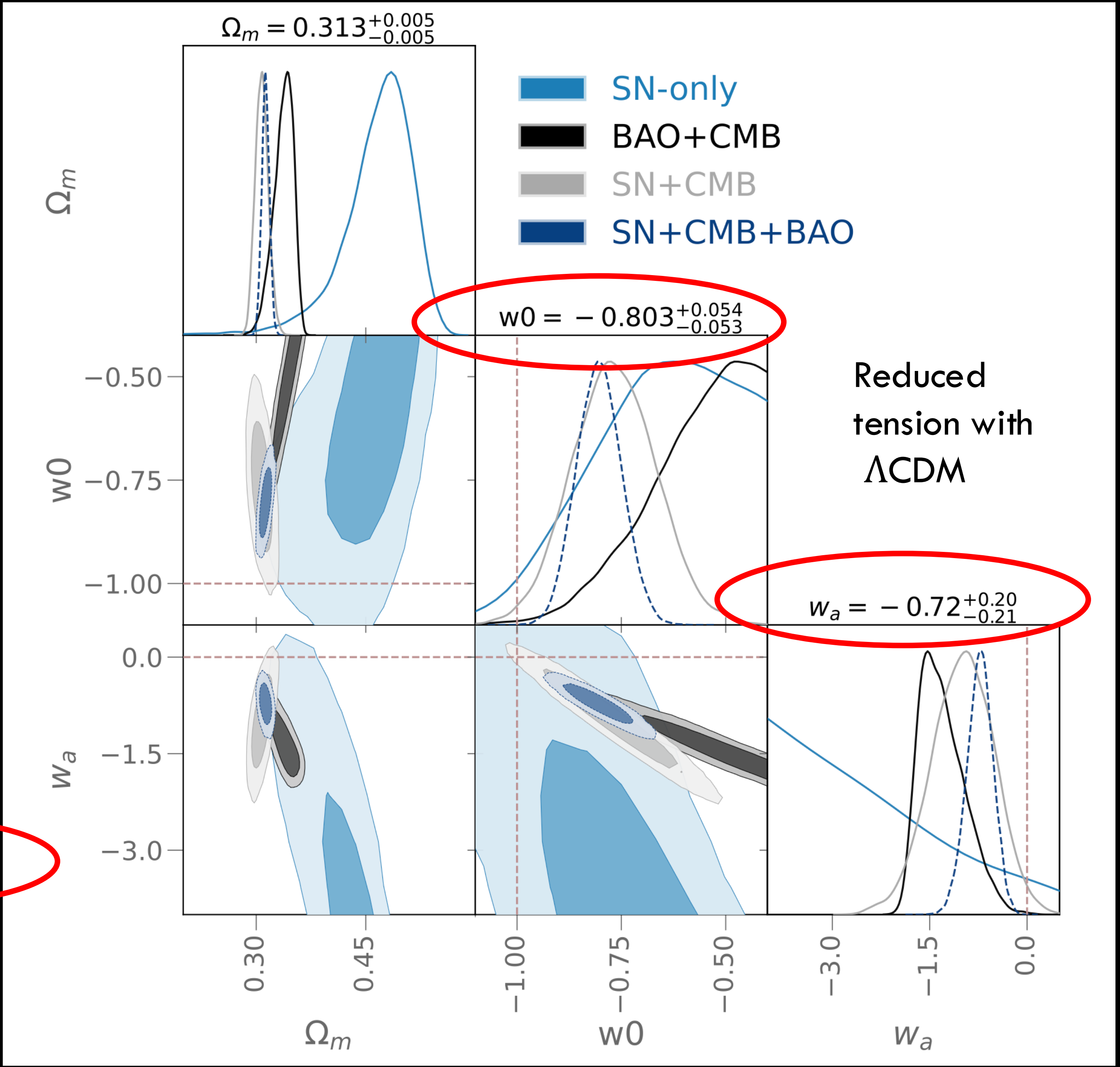
DES SN+DESI DR2 BAO



4.2σ from ΛCDM

DESI DR2, March 2025

DES-Dovekie: DES SN recalibration



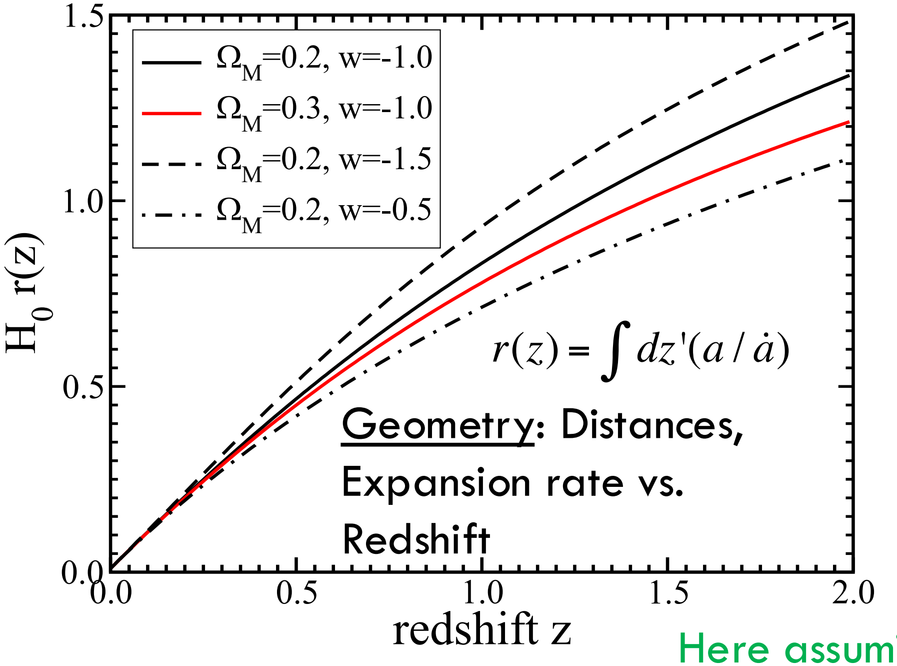
Reduced tension with ΛCDM

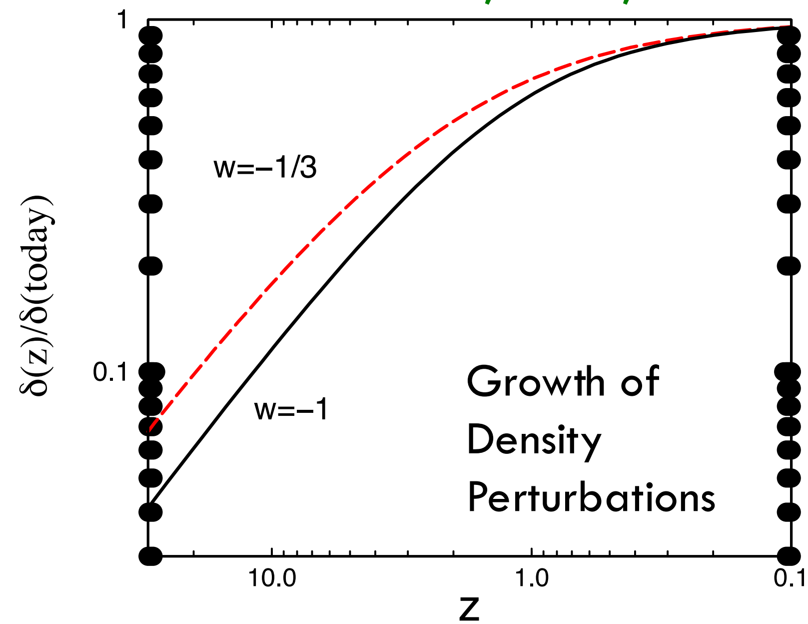
Popovich+ Nov. '25

3.2σ from ΛCDM

Probes of Dark Energy

JF, Turner, Huterer '08





Here assuming w =constant

Expansion History

Type Ia Supernovae

Baryon Acoustic Oscillations

Growth of Structure

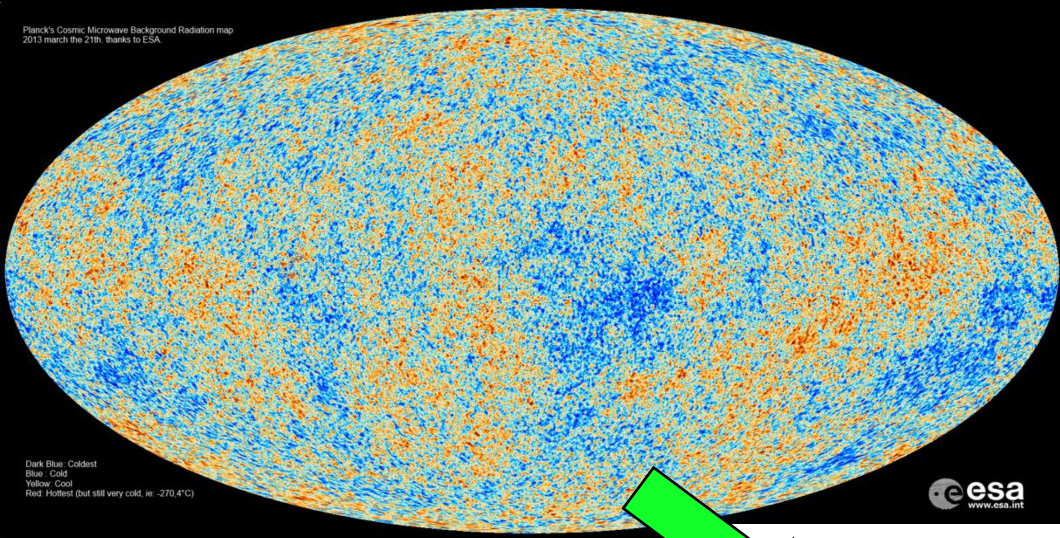
Weak Lensing

Galaxy Clustering

Cluster Abundance

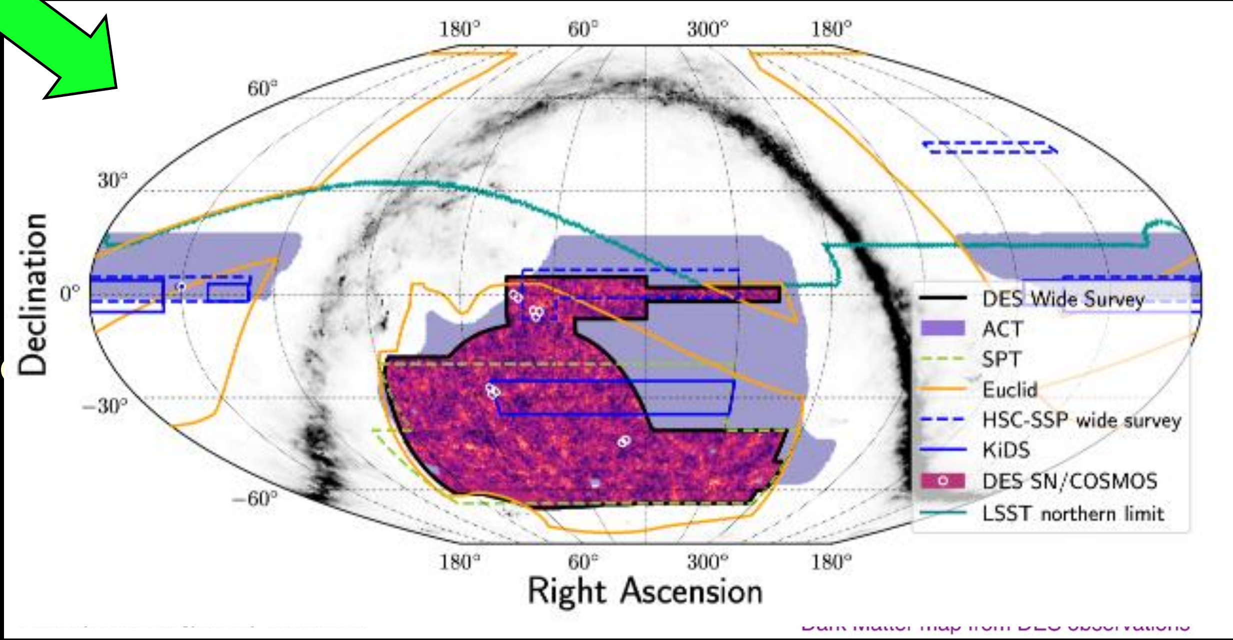
Redshift Space Distortions

Growth of Structure

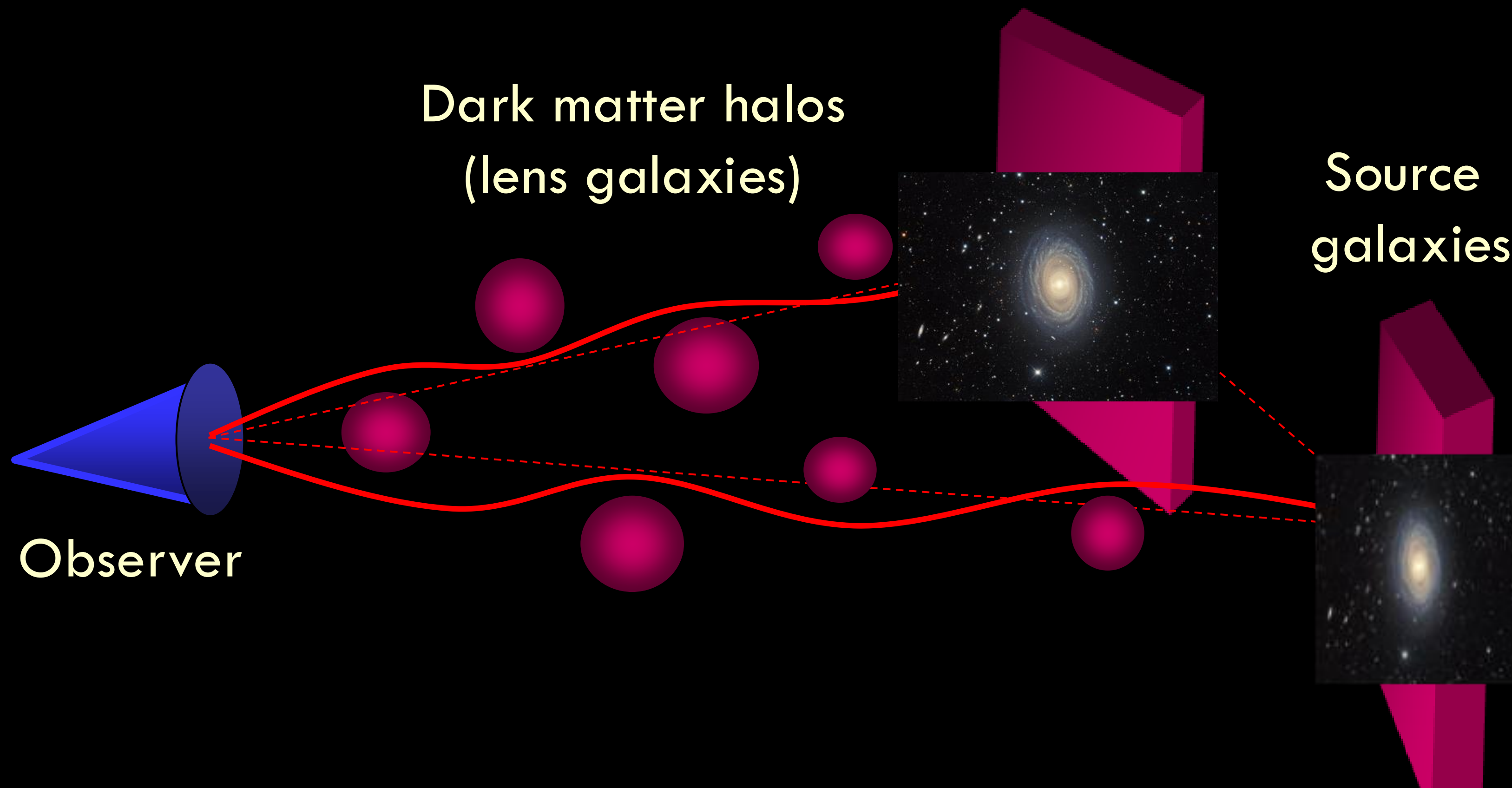


Planck Cosmic Microwave Background (CMB) map of temperature anisotropy at $z_{*} \sim 1000$.

Dark Energy Survey weak lensing map of large-scale structure today. Perturbations have grown by factor 10^3 on large scales.



Weak Lensing and Clustering



Cosmic shear: ~1% correlated distortions of source galaxy shapes

Galaxy-galaxy lensing: Foreground lens galaxy positions correlate with distorted source galaxy shapes

Galaxy clustering: correlate foreground lens galaxy positions with each other

DES Y6 3x2pt Results vs ΛCDM

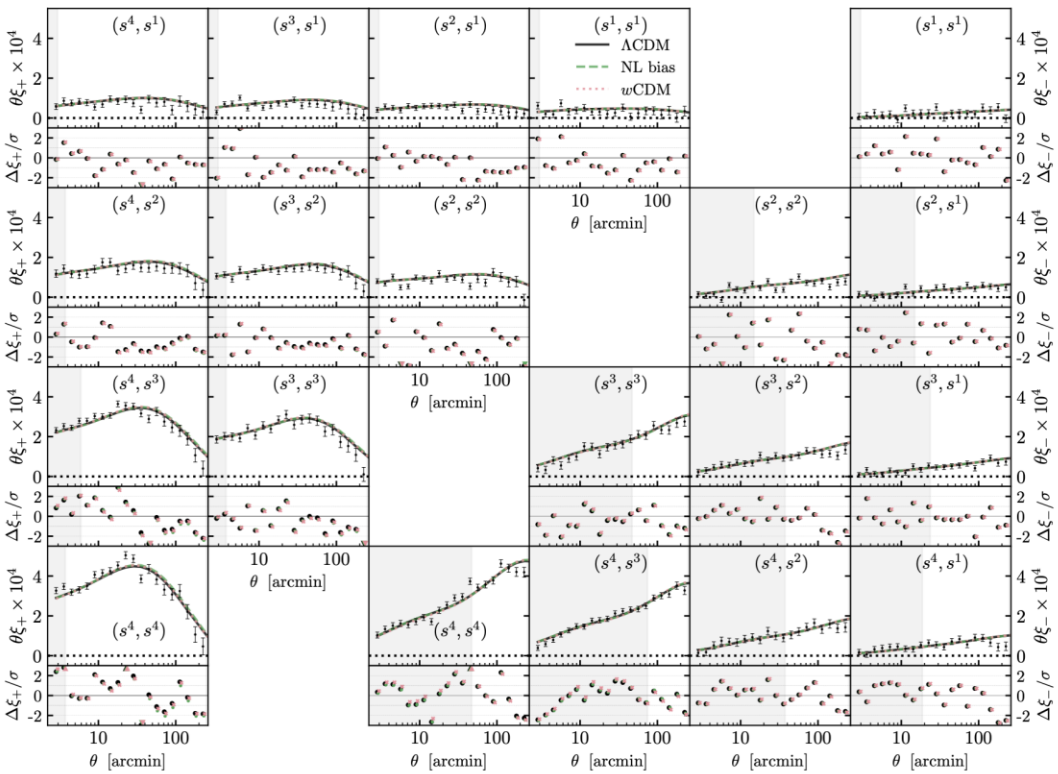
DES Collaboration 2026

Measurements+joint model fit to ΛCDM

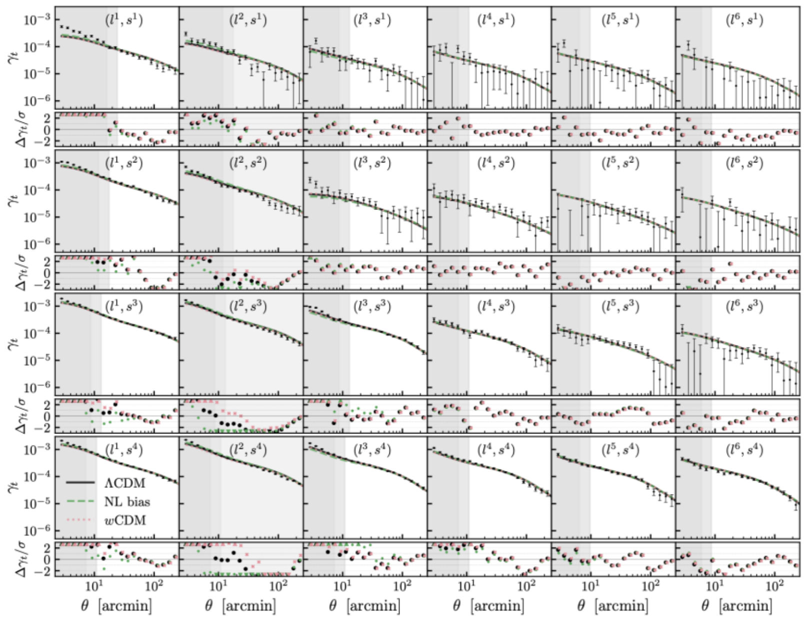
140M source galaxies, 9M lens galaxies

+45/51 nuisance parameters

cosmic shear 4 source galaxy photo-z bins

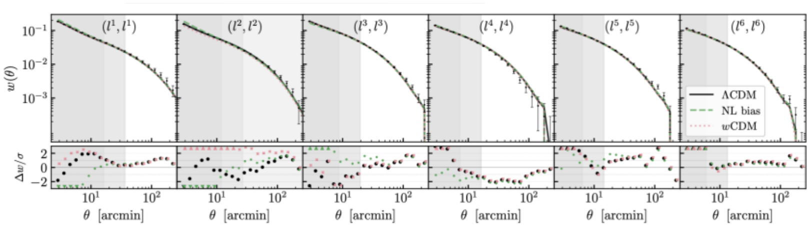


galaxy-galaxy lensing



6 lens galaxy photo-z bins

galaxy clustering

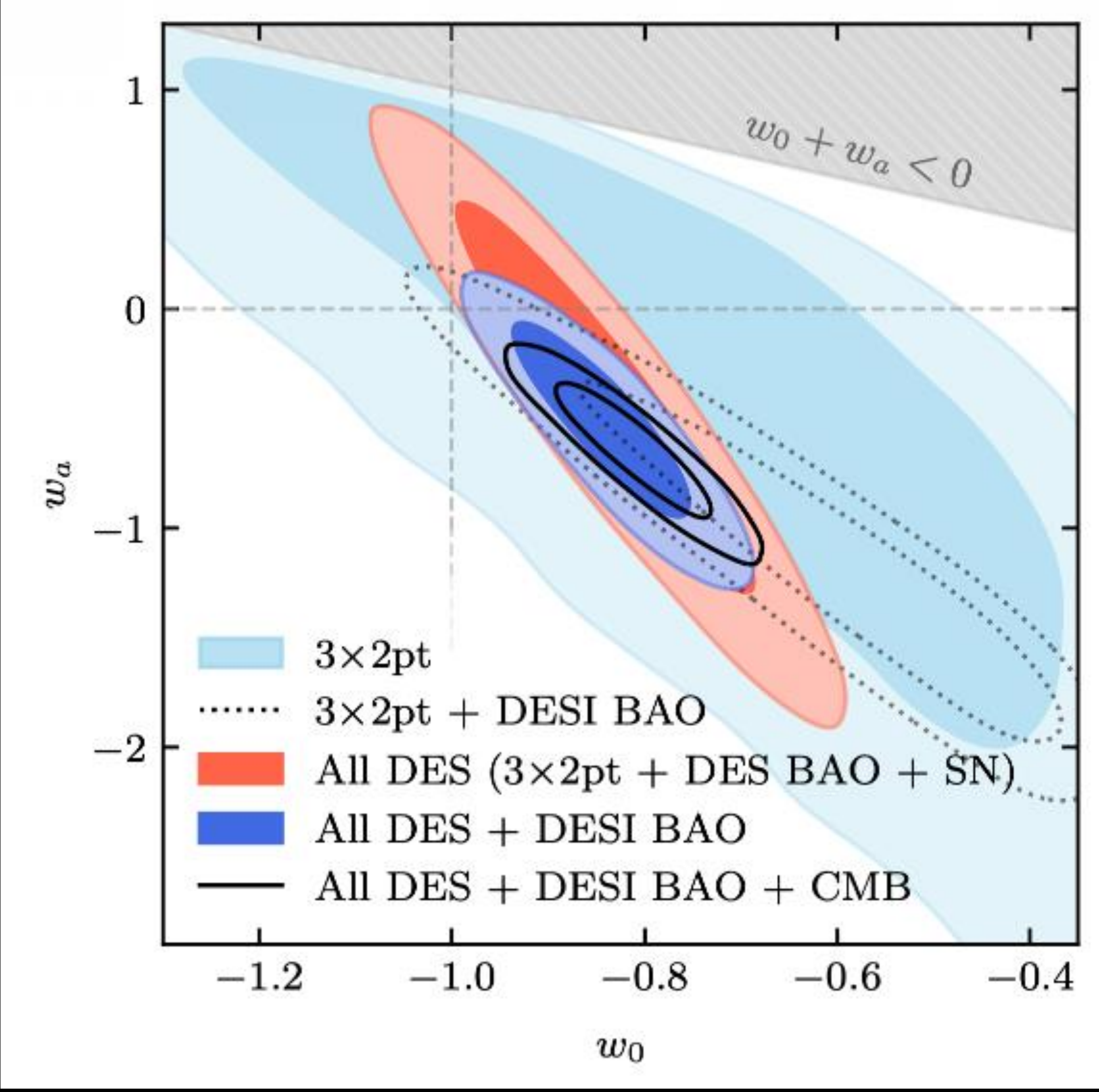


DES Y6 Results vs $w_0 w_a$CDM

DES Collaboration 2026

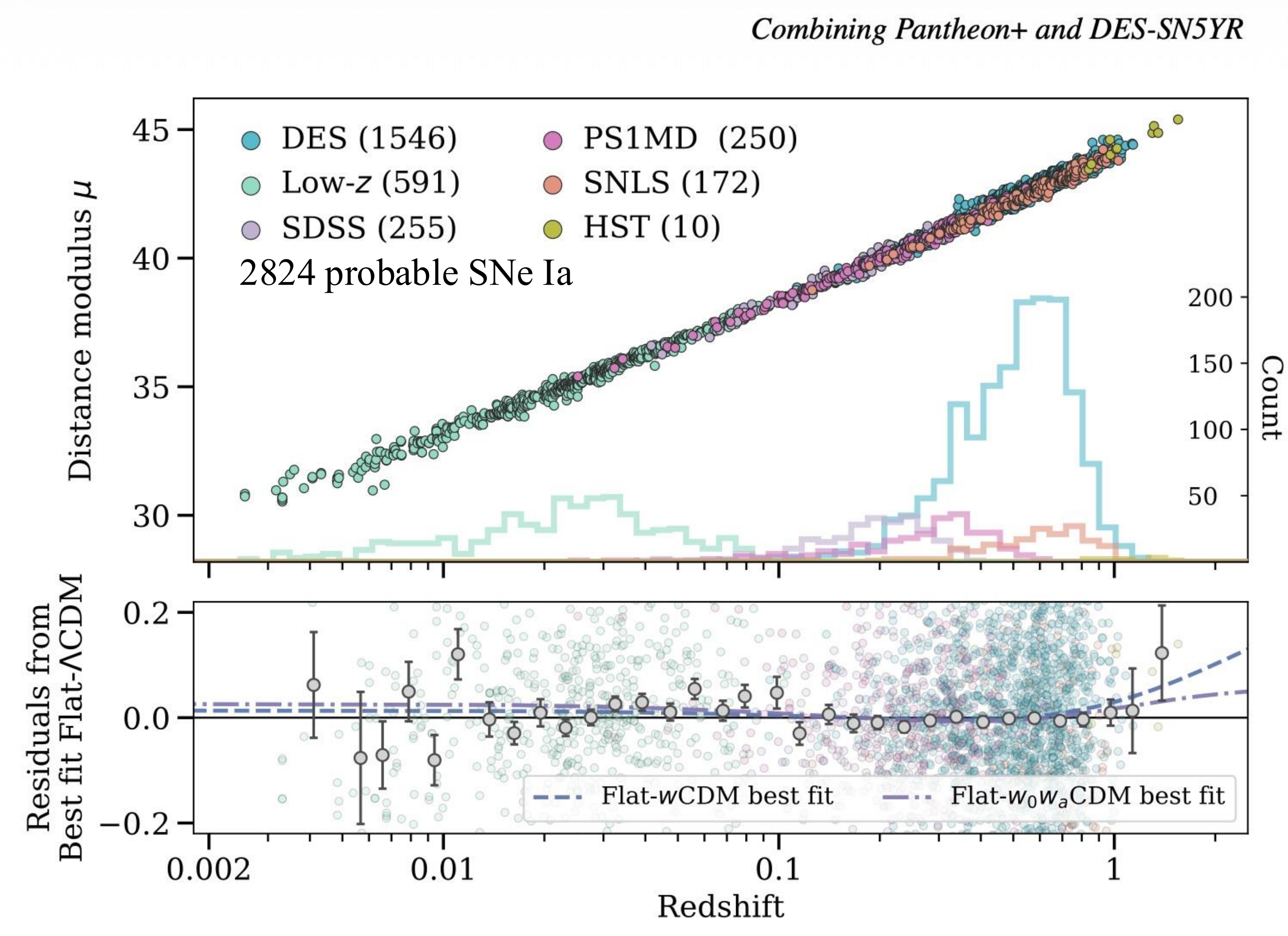
$$\left.\begin{array}{l} w_0 = -0.82 \pm 0.05 \\ w_a = -0.63^{+0.21}_{-0.18} \end{array}\right\}$$ All DES + DESI BAO + CMB (68% C.L.),

3.0σ away from ΛCDM



SN-UNITE: Pantheon+ + DES SN

Camilleri+ July '26

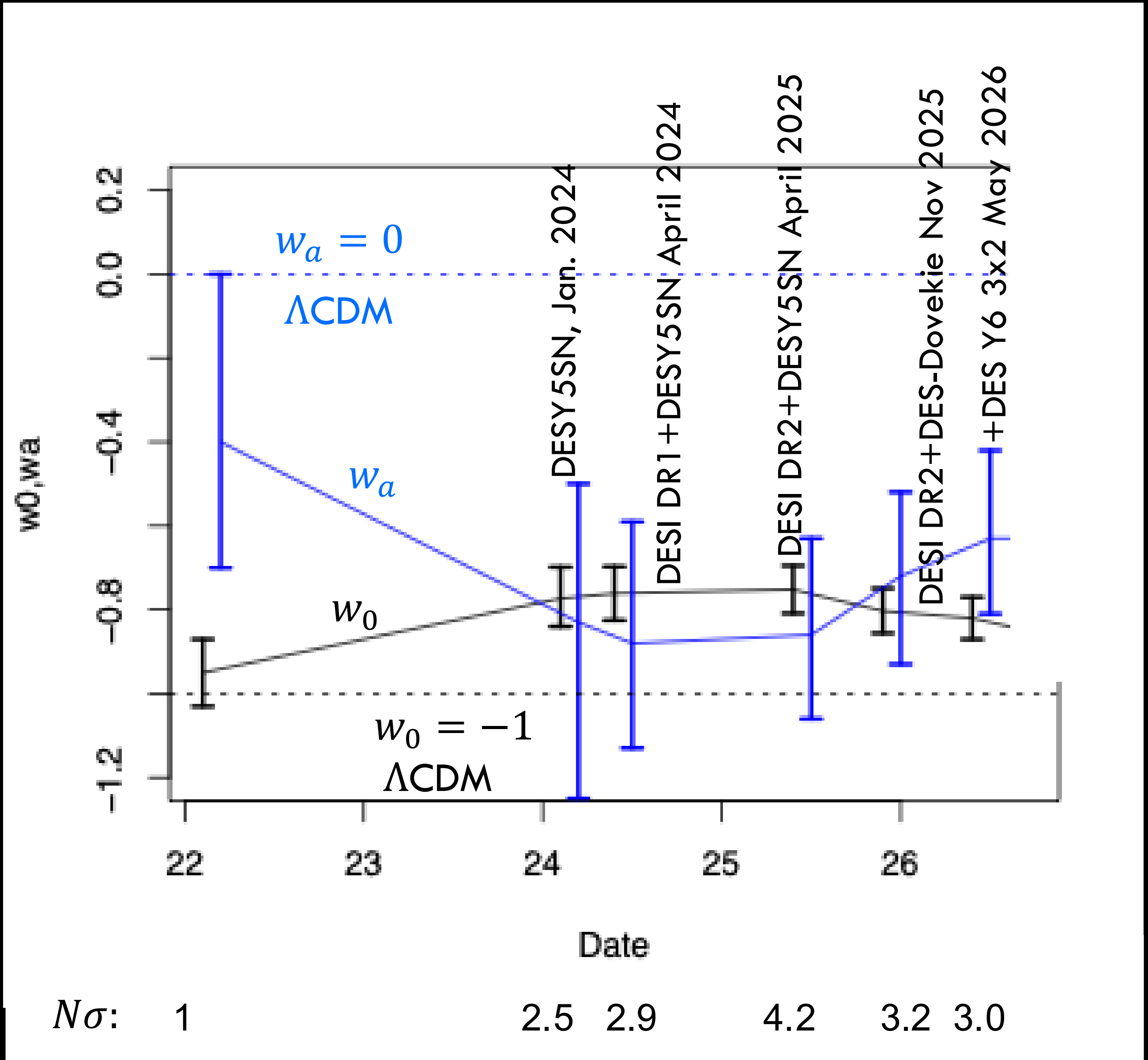


Constraints over time: Huchra-Freedman plot

Combined constraints including CMB

Degeneracies anti-correlate w_0, w_a

At 1-σ, we're no farther from ΛCDM than we were in Jan. 2024 with DES SN results (pre-DESI).



What is the Nature of Dark Energy?

- To constrain the *physical* nature of dark energy, use the data to test *physical** models of dark energy.
  - *e.g., derived from a technically natural, fundamental or Effective Field Theory Lagrangian
- The CMB community has followed this approach in constraining models of primordial inflation.
- Ockham's Razor: start with simple (few-parameter) models and only add complexity as demanded by the data:
  - ΛCDM: effectively 1 parameter (Ω_m)
  - $w_0 w_a$CDM: 3 parameters, no physics connection
  - w_ϕCDM: 2 parameters, fit to Lagrangian models

(+ amplitude of perturbations)

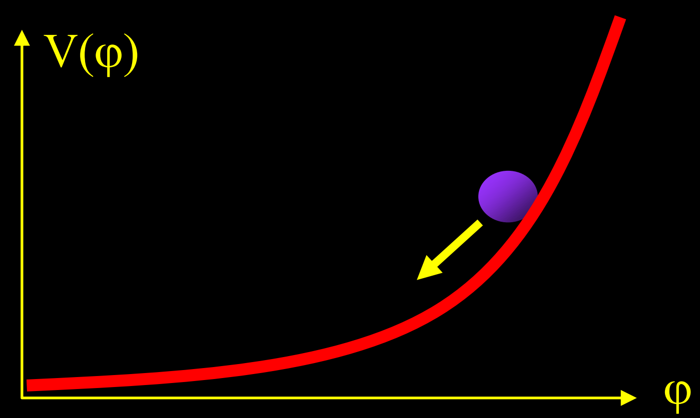
Scalar Field Dark Energy

- Evolving Dark Energy due to a very light scalar field $\varphi(x,t)$, slowly rolling in a potential $V(\varphi)$:

$$\ddot{\varphi} + 3\mathrm{H}\dot{\varphi} + \frac{\mathrm{dV}}{\mathrm{d}\varphi} = 0$$

- Density & pressure:

$$\rho = \tfrac{1}{2}\dot{\varphi}^2 + V(\varphi)$$
$$P = \tfrac{1}{2}\dot{\varphi}^2 - V(\varphi)$$



- Slow roll:

$$\tfrac{1}{2}\dot{\varphi}^2 < V(\varphi) \Rightarrow P < 0 \Leftrightarrow w < 0 \text{ and time - dependent}$$

Simplest Evolving DE Model: Free, Massive Scalar

Damped Harmonic Oscillator:

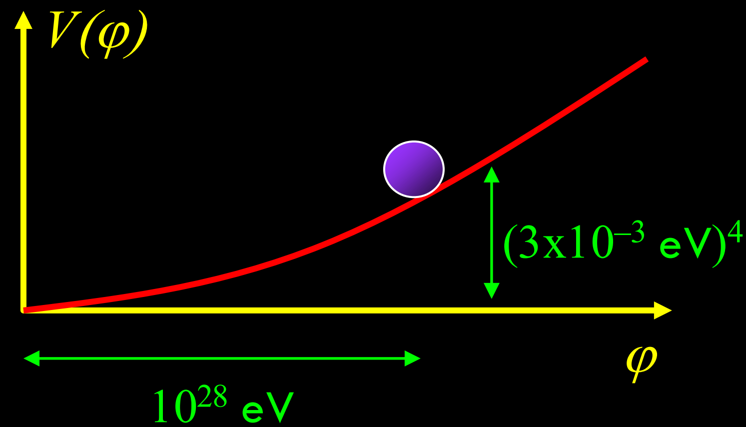
$$\ddot{\varphi} + 3H\dot{\varphi} + \frac{dV}{d\varphi} = 0$$

$$V = \frac{1}{2} m^2 \varphi^2$$

Field frozen until $H(t) \sim m$:
`viscosity' falls as expansion slows.
For slow roll (acceleration):
$m \lesssim 3H_0 \sim 10^{-33}$ eV.

$$H_0^2 \sim \frac{V}{M_{Pl}^2} \sim m^2 \left(\frac{\varphi^2}{M_{Pl}^2}\right) \sim H_0^2 \left(\frac{\varphi^2}{M_{Pl}^2}\right)$$

$$\rightarrow \varphi \sim M_{Pl} \sim 10^{28}\text{eV}$$



Ultra-light particle: expect Scalar Dark Energy nearly homogeneous.
Thawing: $w > -1$ and evolves in time from $w = -1$ (obeys NEC).
Naturalness: how do we make ultra-light scalars in particle physics?

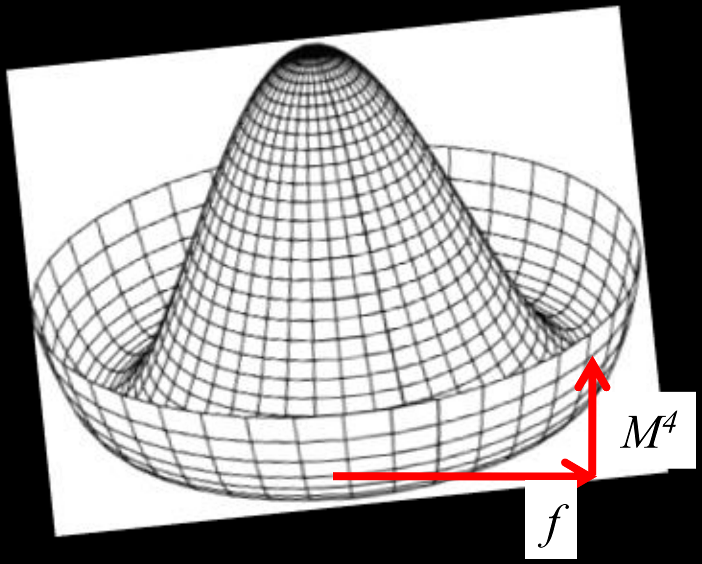
Naturalness: PNGB* Model of Dark Energy
(aka Ultra-light Axion)

*pseudo-Nambu-Goldstone Boson

$$V(\phi) = M^4\left(1+\cos(\phi/f)\right)$$

$$M \sim 10^{-3}\,\mathrm{eV},\ f \sim M_{Pl} \sim 10^{28}\,\mathrm{eV}$$
$$\rightarrow m_\varphi \sim \frac{M^2}{f} \sim 10^{-33}\,\mathrm{eV}$$

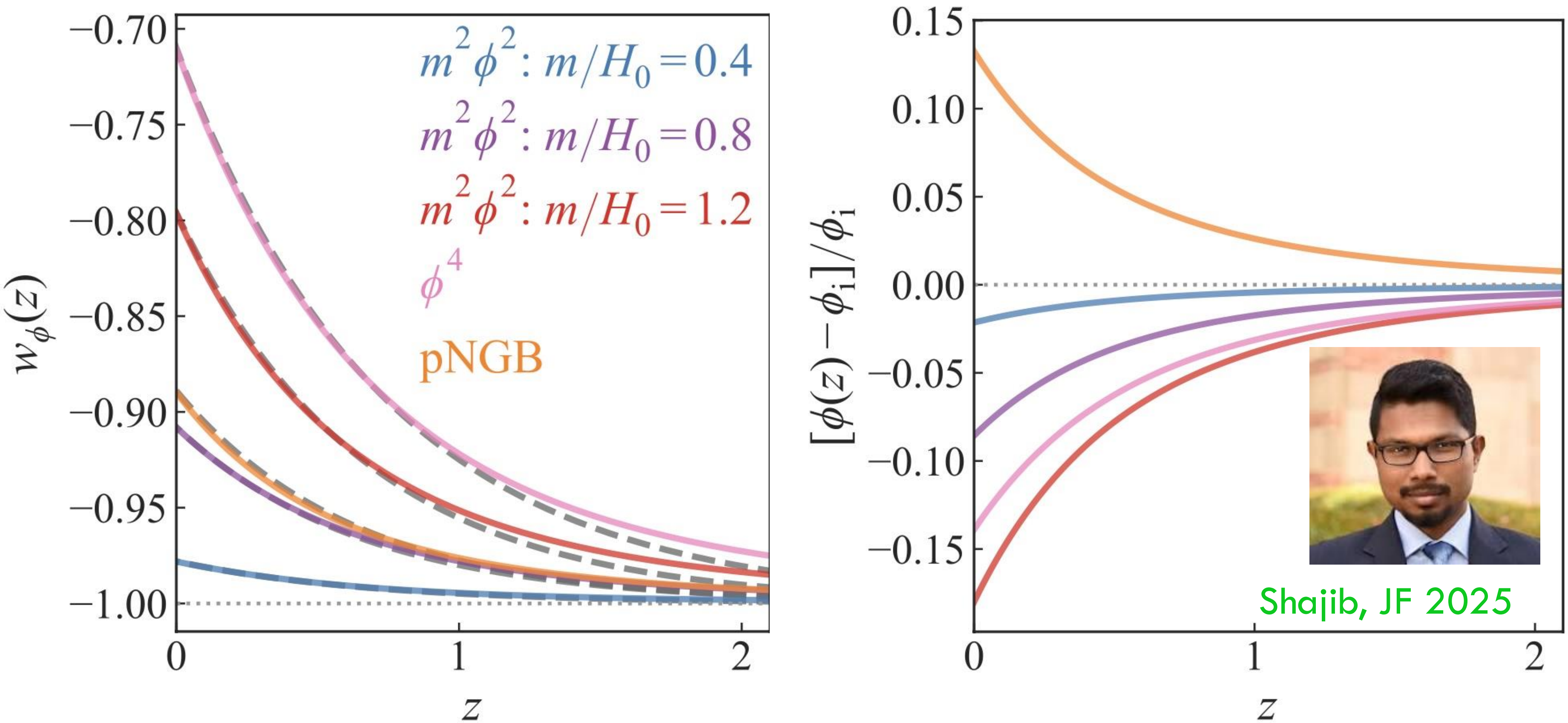
JF, Hill, Stebbins, Waga 1995



- Spontaneous U(1) symmetry breaking at fundamental scale $f \sim M_{Pl}$.
- Explicit breaking at much lower scale $M \ll f$.
- Hierarchy between M and f and thus m_φ protected by shift symmetry.
- Tiny hierarchy $\frac{m_\phi}{\phi} \sim \frac{H_0}{M_{Pl}} \sim \frac{10^{-33}\mathrm{eV}}{10^{28}\mathrm{eV}} \sim 10^{-61}$ is technically natural.
- (For the Higgs field, $\frac{m_\phi}{\phi} \sim 1$. For QCD axion, $\frac{m_\phi}{\phi} \sim 10^{-14}$.)

Thawing Scalar Field Models: w_ϕCDM

Broad class of simple models display quasi-universal behavior and are described by a *single* parameter beyond ΛCDM: w_0



When field thaws, $m \sim H$, it rolls on the Hubble timescale

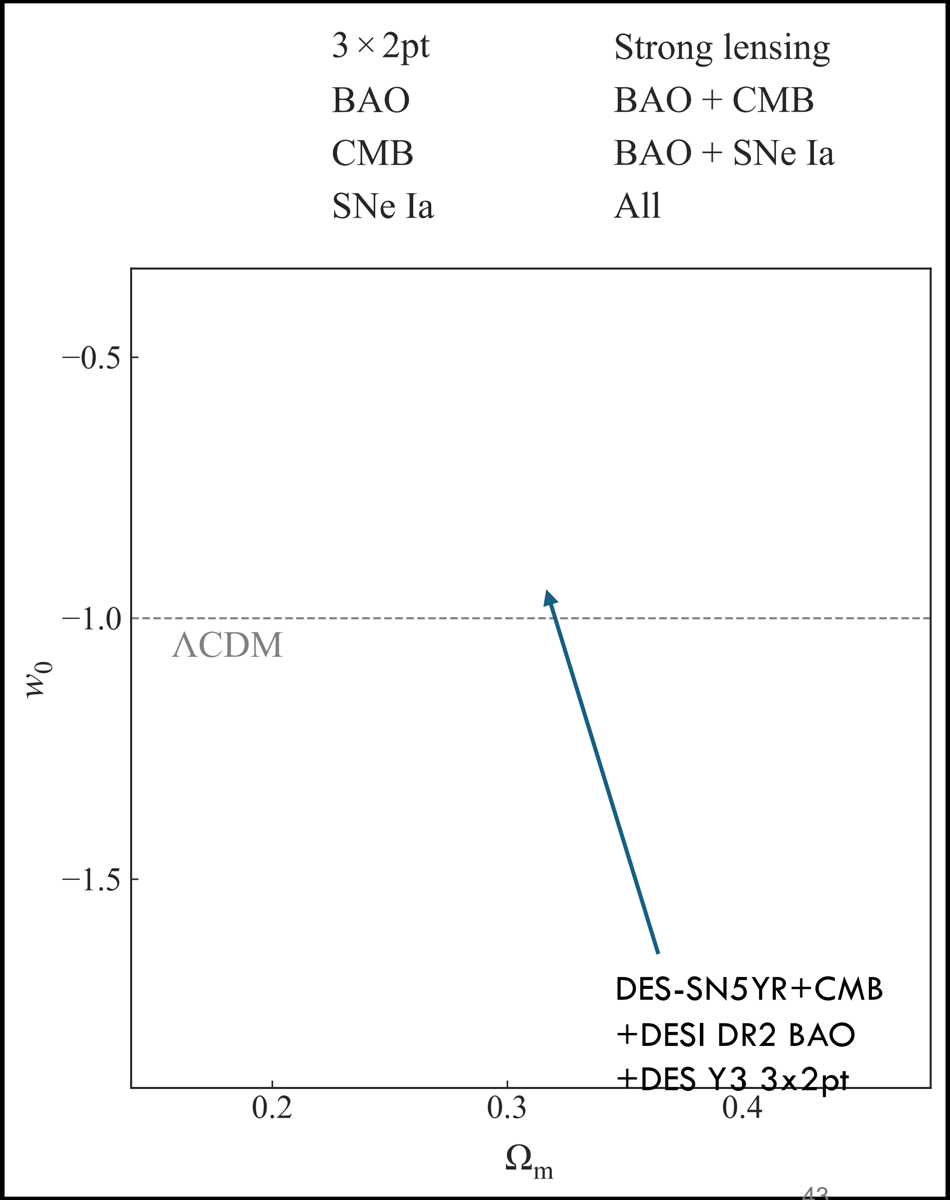
$w(z) \cong -1 + (1 + w_0)e^{-1.5z}$ where $w_0 \equiv w(z = 0)$

Observational Constraints on w_ϕCDM Models

- Best fit parameters:

$$w_0 = -0.904 \pm 0.034$$
$$\Omega_m = 0.314 \pm 0.005$$

- Best-fit scalar field mass $m = 0.8H_0 = 1.1 \times 10^{-33}$ eV, φ_i/M_{Pl}=0.5.
- 2.9σ from ΛCDM:
  reduces BAO-SN tension,
  increases BAO-CMB tension
- Favored by Bayesian evidence ratio over ΛCDM.



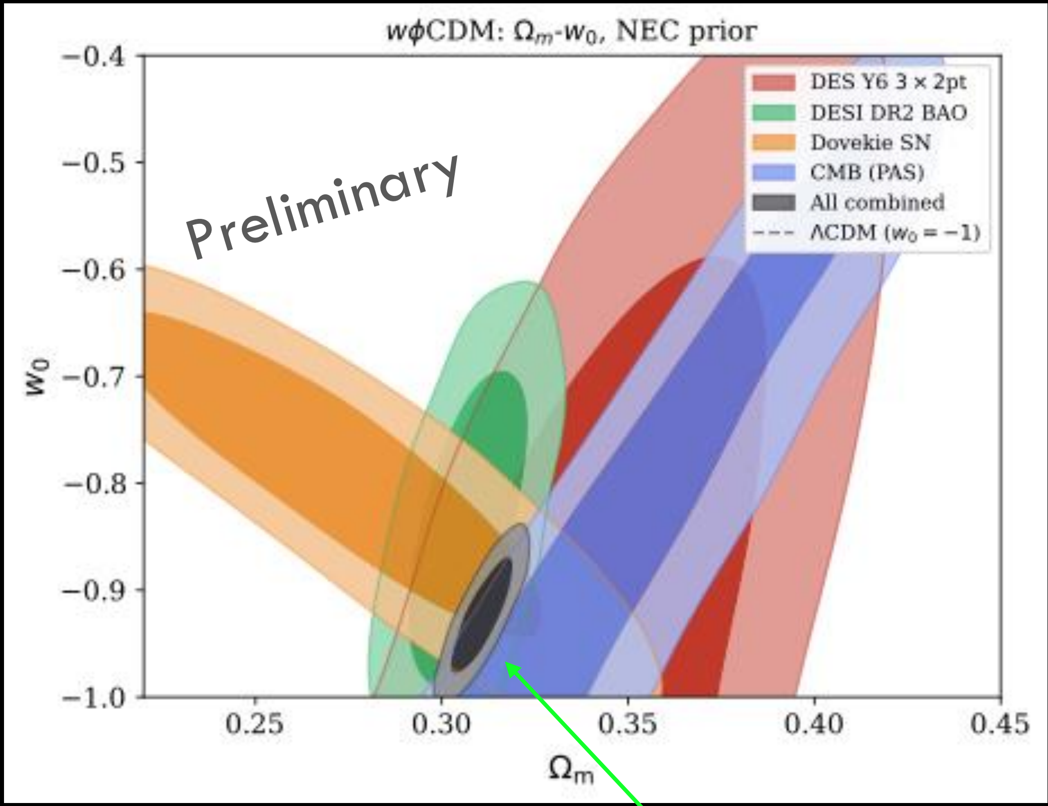
See also Camilleri+ 2024, Lodha+2024, 2025, Urena-Lopez+2025

Shajib, JF 2025

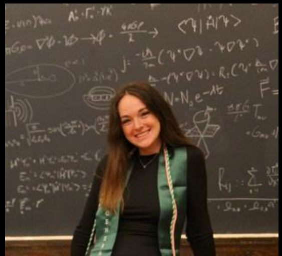
Updated Constraints on w_ϕCDM Models

- Best fit parameters:

$$w_0 = -0.922 \pm 0.035$$
$$\Omega_m = 0.311 \pm 0.005$$



- 2.2σ from ΛCDM;
- Bayesian evidence ratio barely/marginally favors over ΛCDM.
- Compare to 3.0σ for w_0, w_a for same data set combination, but with one more parameter.

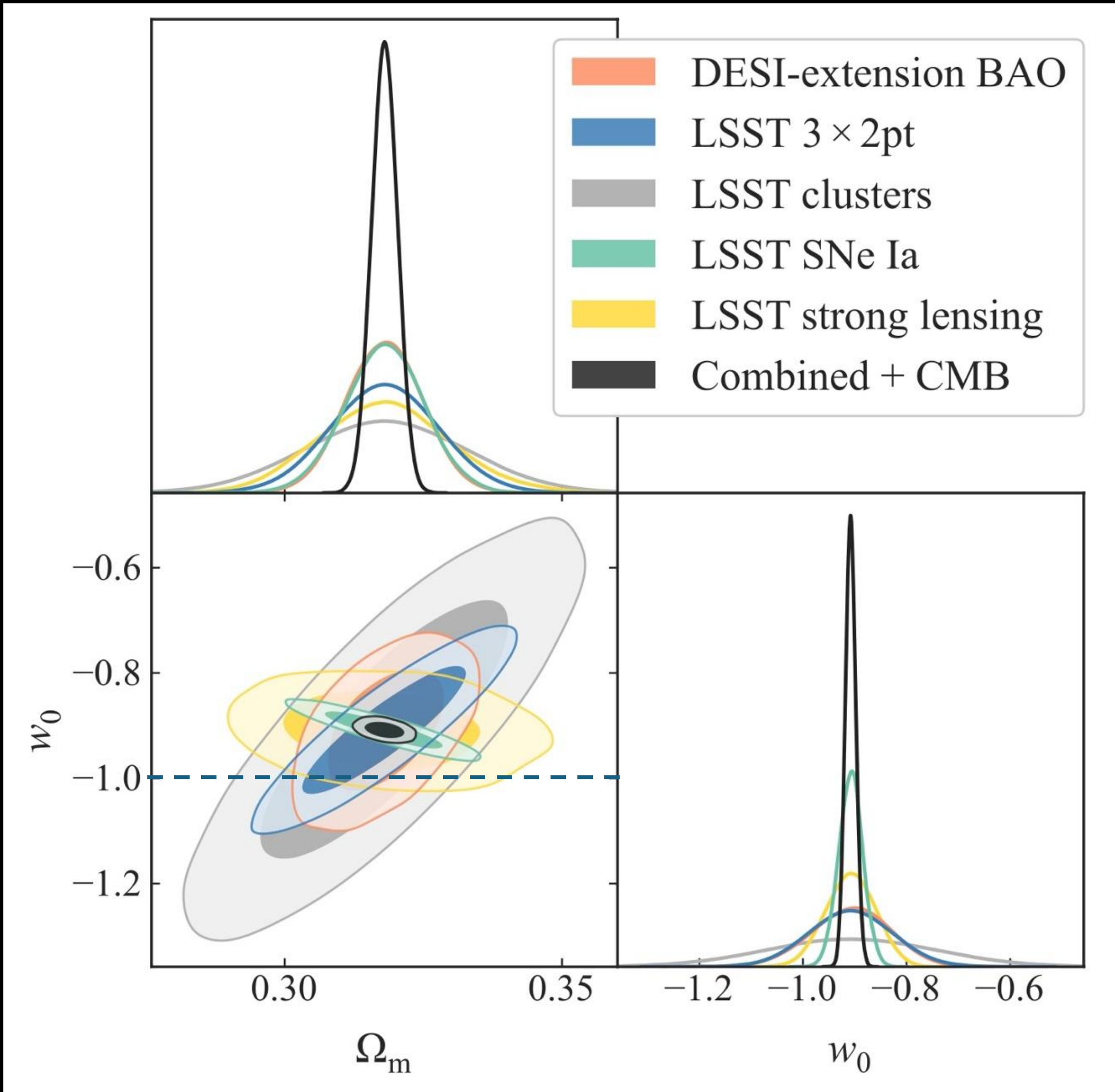


DES-Dovekie+CMB +DESI DR2 BAO +DES Y6 3x2pt

Jones, JF, Shajib 2026

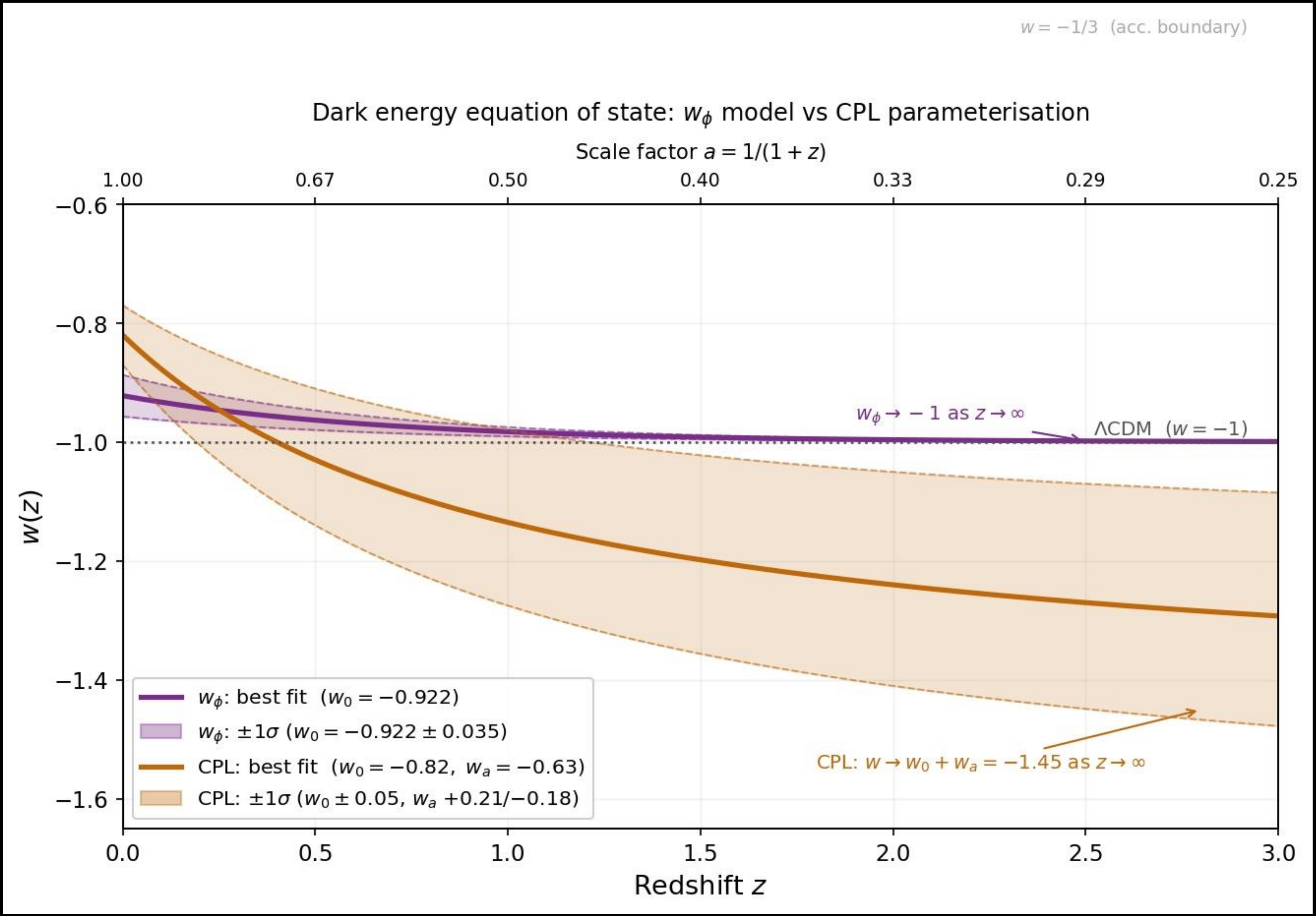
Testing w_ϕCDM

- Forecast constraints for DESI extension plus Rubin LSST.
- $\sigma(w_0)$ reduced from 0.035 to 0.01.
- Current best-fit w_ϕCDM model will be $\sim 8\sigma$ from ΛCDM.

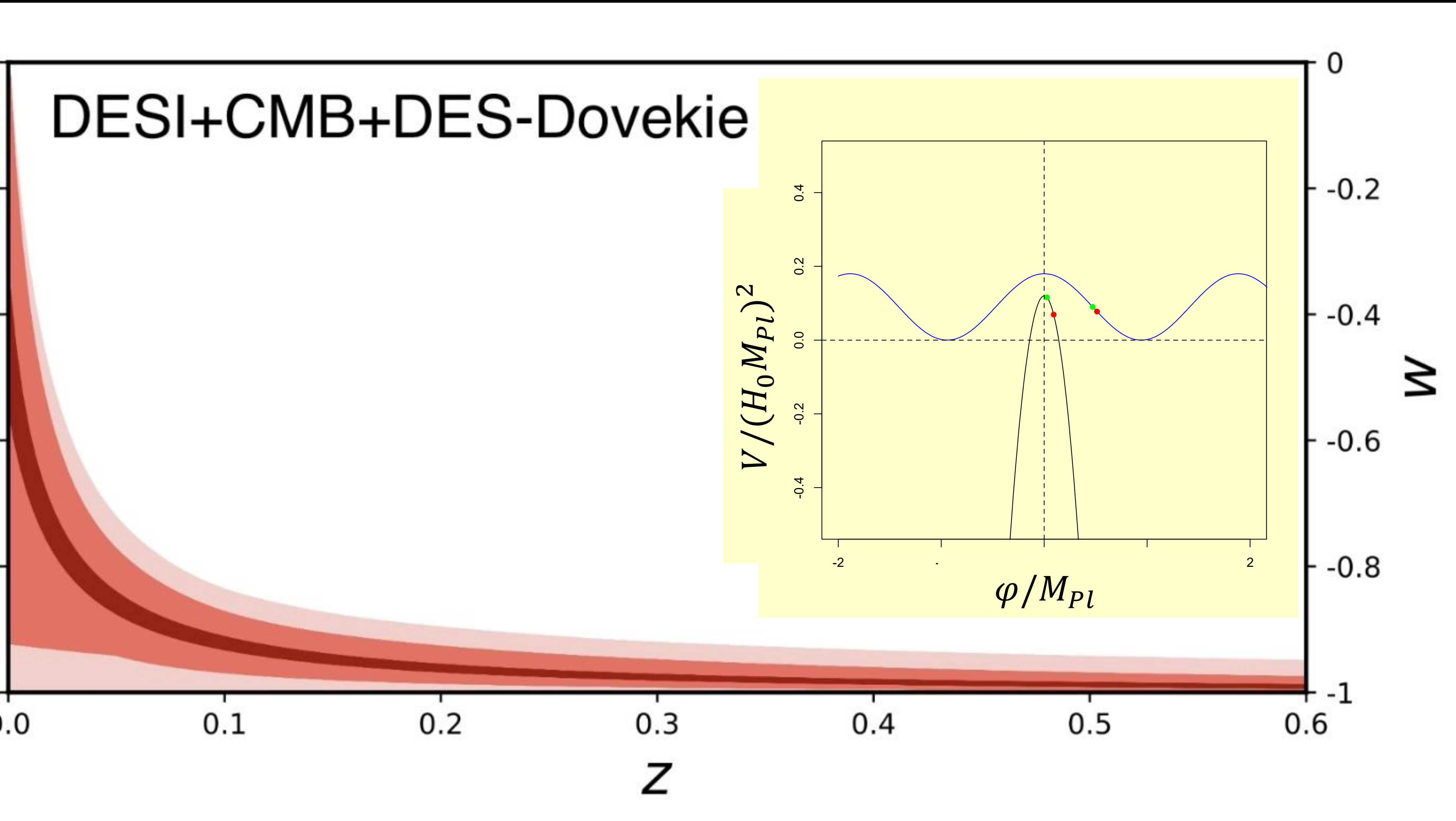


Shajib, JF 2025

Current Best-fit w_ϕCDM and $w_0 w_a$ Models



Broadening the Theory Prior:
Scalar Models that can evolve faster



Shlivko 2025

Is Dark Energy Evolving?

- Evolving DE models fit data better than ΛCDM, but to a degree that only *modestly* or marginally justifies added model complexity.
- Evolving DE remains a hint, not a discovery.
- The evidence for phantom-crossing ($w < -1$ at early times) is not yet compelling (to me).
- If the current hints for evolving DE are real, DESI+Rubin LSST +Euclid+Roman should nail it definitively.
- Geometric probes currently have strongest DE constraining power (BAO, SN), but structure growth becoming competitive and is complementary.
- Standard $w_0 w_a$ prescription not physically motivated: we should test physical or physically-inspired DE models (as we do for inflation).
- w_ϕCDM models are simplest example: ultra-light scalar (e.g., axion) with mass $m \sim H_0$ fits current data marginally better than ΛCDM.

AI won't take over until it's funnier than we are

George Efstathiou and Andrew Jaffe are at a cosmology conference.

George stands up and declares, "The Hubble tension is not a crisis. The systematics are in the local measurements."

Andrew nods thoughtfully, opens his laptop, updates his blog, and writes: *"George may be right. Then again, he may not be. The posterior is broad."*

George reads it and says, "Andrew, that's not a position."

Andrew says, "On the contrary — it's a perfectly normalised one."

--Claude

AI won’t take over until it’s funnier than we are

George Efstathiou and Andrew Jaffe are at a cosmology conference.

George stands up and declares, "The Hubble tension is not a crisis. The systematics are in the local measurements."

Andrew nods thoughtfully, opens his laptop, updates his blog, and writes: *"George may be right. Then again, he may not be. The posterior is broad."*

George reads it and says, "Andrew, that's not a position."

Andrew says, "On the contrary — it's a perfectly normalised one."

--Claude

Claude knows that Andrew is a Bayesian and has a blog.
Claude knows that George is George.

The Real Cosmic Tension Question for Andrew

Beatles or Stones?

(Claude would not give me Bayesian odds on Andrew’s view.)

Happy Birthday Andrew!

The Random Universe: a meeting to celebrate Andrew Jaffe at 60

Event | **Conference**

29 June 2026 - 2 July 2026

14.00 - 13.00 BST

Lecture Theatre 2, Blackett Building
South Kensington Campus

Accessibility information

Audience: Invitees only
Cost: Free
Tickets: Tickets to be purchased in advance

Register now

A conference to celebrate Andrew Jaffe's many contributions to cosmology and the exciting scientific developments still to come on his 60th birthday.